Notes on Linear Algebra

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1 Vector Spaces

1.1 Fields

We will use $\mathbf{F}$ to denote the sets $\mathbb{R}$ and $\mathbb{C}$. The complex numbers have the following properties:

1. commutativity: $\alpha + \beta = \beta + \alpha$ and $\alpha\beta = \beta\alpha$ for all $\alpha, \beta \in \mathbb{C}$.
2. associativity: $(\alpha + \beta) + \lambda = \alpha + (\beta + \lambda)$ and $(\alpha\beta)\lambda = \alpha(\beta\lambda)$ for all $\alpha, \beta, \lambda \in \mathbb{C}$.
3. identities: $\lambda + 0 = \lambda$ and $\lambda \cdot 1 = \lambda$ for all $\lambda \in \mathbb{C}$.
4. additive inverse: For every $\alpha \in \mathbb{C}$ there exists a unique $\beta \in \mathbb{C}$ such that $\alpha + \beta = 0$.
5. multiplicative inverse: For every $\alpha \in \mathbb{C}$ with $\alpha \neq 0$ there exists a unique $\beta \in \mathbb{C}$ such that $\alpha\beta = 1$.
6. distributive property: $\lambda(\alpha + \beta) = \lambda\alpha + \lambda\beta$ for all $\lambda, \alpha, \beta \in \mathbb{C}$.

Definition 1.1 (Field)**.** A *field* is a set containing at least two distinct elements called 0 and 1, along with the operations of addition and multiplication satisfying the properties listed previously.

1.2 Vector Spaces

Definition 1.2 (Vector Space)**.** A *vector space* is a set V with an addition on V and a multiplication on V that satisfies the following properties:

1. Commutativity: $u + v = v + u$ for all $u, v \in V$.
2. Associativity: $(u+v)+w = u+(v+w)$ and $a(bv) = (ab)v$ for all $u, v, w \in V$ and $a, b \in \mathbf{F}$.
3. Additive Identity: There exists an element $0 \in V$ such that $v + 0 = v$ for all $v \in V$.
4. Additive Inverse: For every $v \in V$, there exists $w \in V$ so that $v + w = 0$.
5. Multiplicative Identity: $1v = v$ for all $v \in V$.
6. Distributive Property: $a(u + v) = au + av$ and $(a + b)v = av + bv$ for all $a, b \in \mathbf{F}$ and for all $u, v \in V$.

Here we list some theorems related to vector spaces. These can be proved using the properties above.

Theorem 1.1 (unique additive inverse)**.** *Every element in a vector space has a unique additive inverse. If $v \in V$, we denote the unique additive inverse $-v$.*

Theorem 1.2 (the number 0 times a vector)**.** $0v = 0$ *for every* $v \in V$.

Theorem 1.3 (a number times the vector 0)**.** $a0 = 0$ *for every* $a \in \mathbf{F}$.

Theorem 1.4 (the number -1 times a vector)**.** $(-1)v = -v$ *for every* $v \in V$.

Theorem 1.5 (unique additive identity)**.** *A vector space has a unique additive identity.*

1.3 Subspaces

Definition 1.3. A subset U of V is called a *subspace* of V if U is also a vector space with the same additive identity, addition, and scalar multiplication as V.

Theorem 1.6. *A subset* U *of* V *is a subspace of* V *if and only if* U *satisfies the following three conditions:*

(i) ***additive identity****:* $0 \in U$

(ii) ***closed under addition****:* $u, w \in U \implies u + w \in U$

(iii) ***closed under scalar multiplication****:* $a \in \mathbf{F}$ *and* $u \in U$ *implies* $au \in U$

Proof.

1. U is a subspace of $V \implies$ (i), (ii), and (iii): If U is a subspace of V, then by Definition 1.3 the additive identity, 0, of U is the same as that of V, so (i) is satisfied. Since U is a vector space, it is closed under addition, so (ii) is satisfied. Since U is a vector space, it is closed under scalar multiplication, so (iii) is satisfied.

2. (i), (ii), and (iii) $\implies$ U is a subspace of V: Since $0 \in U$ and $U \subset V$, $0 \in V$, so U has the same additive identity as V. By (ii) and (iii), addition on U and scalar multiplication on U is well-defined. Since addition and scalar multiplication on V satisfy commutativity and associativity, U satisfies commutativity and associativity with respect to the same addition and multiplication. The distributive property also holds on U for the same reason. For each element $u \in U$, $-u = (-1)u \in U$, so every element of U has an additive inverse. Since $1v = v$ for all $v \in V$, $1u = u$ for all $u \in U$ since $U \subset V$.

□

Here are some examples of subspaces. In all these cases, we use Theorem 1.6 to show all of them are subspaces:

Example 1.1.

(a) If $b \in \mathbf{F}$, then:

$$S = \{(x_1, x_2, x_3, x_4) \in \mathbb{F}^4 : x_3 = 5x_4 + b\}$$

is a subspace of $\mathbf{F}^4$ if and only if $b = 0$.

Proof. Note that $(0,0,0,0) \in S$ if and only if $0 = 5 \cdot 0 + b$. That is, $(0,0,0,0) \in S$ if and only if $b = 0$. Suppose $x = (x_1, x_2, x_3, x_4) \in S$ and $y = (y_1, y_2, y_3, y_4) \in S$. Then, $x_3 = 5x_4 + b$ and $y_3 = 5y_4 + b$. Then, $x_3 + y_3 = 5(x_4 + y_4) + 2b$. So, $x + y \in S$ if and only if $b = 0$. For scalar multiplication: let $\lambda \in \mathbf{F}$. Since x satisfies $x_3 = 5x_4 + b$, it follows $\lambda x_3 + 5\lambda x_4 + \lambda b$. Then, $\lambda x \in S$ if and only if $b = 0$. Therefore, S is a subspace iff $b = 0$. □

(b) The set of continuous real-valued functions on the interval $[0, 1]$ is a subspace of $\mathbb{R}^{[0,1]}$.

Proof. The function $z : [0, 1] \to \mathbb{R}$ where $z(x) = 0$ for all $x \in [0, 1]$ is the additive identity for $\mathbb{R}^{[0,1]}$, and it is continuous. Let f, g be continuous real-valued functions on the interval $[0, 1]$. Then, $f + g$ is a continuous real-valued function on $[0, 1]$, so the set is closed under addition. Similarly, λf for $\lambda \in \mathbf{F}$ is continuous, so the set is closed under scalar multiplication. Therefore, the set of continuous real-valued functions on the interval $[0, 1]$ is a subspace of $\mathbb{R}^{[0,1]}$. □

(c) The set of differentiable real-valued functions on $\mathbb{R}$ is a subspace of $\mathbf{R}^{\mathbb{R}}$.

Proof. The function $z : \mathbb{R} \to \mathbb{R}$ where $z(x) = 0$ for all $x \in \mathbb{R}$ is differentiable. Also, if f, g are differentiable real-valued functions, then $f + g$ is differentiable. If $\lambda \in \mathbf{F}$, then λf is differentiable. Therefore, the set is closed under addition and scalar multiplication and has the same additive identity as $\mathbb{R}^{\mathbb{R}}$. So, it is a subspace of $\mathbb{R}^{\mathbb{R}}$. □

(d) Consider the zero function $z : (0, 3) \to \mathbb{R}$ where $z(x) = 0$ for all $x \in (0, 3)$. Then, z is differentiable, real-valued, and $z'(2) = b$ if and only if $b = 0$. Let $\lambda \in \mathbb{R}$. Then, $(\lambda f)(x)$ is differentiable whenever f is differentiable. Furthermore, if $f'(2) = b$, then $\lambda f'(2) = \lambda b = b$ if and only if $b = 0$. Let f and g be differentiable real-valued elements of $\mathbb{R}^{(0,3)}$ satisfying $f'(2) = b$ and $g'(2) = b$. Then, $f+g$ is differentiable and $(f+g)'(2) = f'(2)+g'(2) = 2b = b$ if and only if $b = 0$. Therefore, the set is a subspace of $\mathbb{R}^{(0,3)}$ if and only if $b = 0$.

(e) We will use $(z_n)_{n=1}^{\infty}$ to denote a sequence of complex numbers. Let $z_n = 0$ for all $n \in \mathbb{N}$. This is the infinite sequence $(0, 0, 0, 0, ...)$, which clearly has limit 0. Let $(z_n)_{n=1}^{\infty}$ be a complex sequence so that $\lim_{n\to\infty} z_n = 0$. We may write each $z_n = a_n + ib_n$ for $a_n, b_n \in \mathbb{R}$. The limit of the complex sequence is 0 if and only if $\lim_n a_n = 0$ and $\lim_n b_n = 0$. Let $\lambda \in \mathbb{C}$. Put $\lambda = \lambda_1 + i\lambda_2$ for $\lambda_1, \lambda_2 \in \mathbb{C}$. Then:

$$\begin{aligned}\lim_{n\to\infty} \lambda z_n &= \lim_{n\to\infty} (\lambda_1 + i\lambda_2) \cdot (a_n + ib_n) \\ &= \lim_{n\to\infty} ((\lambda_1 a_n - \lambda_2 b_n) + i \cdot (\lambda_1 b_n + \lambda_2 a_n)) \\ &= 0\end{aligned}$$

since $\lim_{n\to\infty} a_n = 0$ and $\lim_{n\to\infty b_n} = 0$. Therefore, $(\lambda z_n)_{n=1}^{\infty}$ has limit zero. Let $(x_n)_{n=1}^{\infty}$ and $(y_n)_{n=1}^{\infty}$ both have limit zero. Then their sum $(x_n + y_n)_{n=1}^{\infty}$ also has limit zero. Therefore, the set of complex sequences with limit zero is a subspace of $\mathbb{C}^{\infty}$.

Definition 1.4. Suppose $V_1, ..., V_m$ are subspaces of V. The *sum* of $V_1, ..., V_m$, denoted by $V_1 + ... + V_m$, is the set of all possible sums of elements of $V_1, ..., V_m$. That is:

$$V_1 + ... + V_m = \{v_1 + ... + v_m : v_1 \in V_1, ..., v_m \in V_m\}$$

Theorem 1.7 (sums of subspaces is the smallest containing subspace)**.** *Suppose $V_1, ..., V_m$ are subspaces of V. Then, $V_1 + ... + V_m$ is the smallest subspace of V containing $V_1, ..., V_m$*

Proof. First, we show V_k is contained in $V_1 + ... + V_m$ for every $k \in [m]$. We can write $v_k = \sum_{j=1}^{m} v_j$ where $v_j = 0$ for all $j \neq k$ and $v_k \in V_k$ is arbitrary. Then, $v_k \in V_1 + ... + V_m$. Therefore, $V_1, ..., V_m$ are contained in $V_1 + ... + V_m$. Suppose U is a subspace of V that contains $V_1, ..., V_m$. For each $k \in [m]$, $v_k \in V_k \implies v_k \in U$. Since U is closed under addition, $\sum_{k=1}^{m} v_k \in U$. Therefore, $V_1 + ... + V_m$ is contained in U. □

1.4 Direct Sums

Definition 1.5. Suppose $V_1, ..., V_m$ are subspaces of V. Then:

(i) The sum $V_1 + ... + V_m$ is called a *direct sum* if each element of $V_1 + ... + V_m$ can be written in only one way as a sum $v_1 + ... + v_m$, where, for each $k \in [m]$, $v_k \in V_m$.

(ii) If $V_1 + ... + V_m$ is a direct sum, then $V_1 \oplus ... \oplus V_m$ denotes $V_1 + ... + V_m$ and indicates that it is a direct sum.

Theorem 1.8 (condition for a direct sum)**.** *Suppose $V_1, ..., V_m$ are subspaces of V. Then, $V_1 + ... + V_m$ is a direct sum if and only if the only way to write 0 as a sum $v_1 + ... + v_m$, where each $v_k \in V_k$, is by taking each v_k equal to 0.*

Proof. We prove the converse first. Suppose the only way to write 0 as a sum $v_1 + ... + v_m$, where each $v_k \in V_k$, is by taking v_k equal to 0 for all $K \in [m]$. Choose an arbitrary $v \in V_1 + ... + V_m$. Then, $v = \sum_{k=1}^{m} v_k$ where $v_k \in V_k$ for each $k \in [m]$. Suppose there exists another set of vectors $\{v'_k\}_{k=1}^{m}$ so that $v = \sum_{k=1}^{m} v'_k$. Then, $\sum_{k=1}^{m} v_k = \sum_{k=1}^{m} v'_k$. That is:

$$\sum_{k=1}^{m} (v_k - v'_k) = 0$$

Since $v_k - v'_k \in V_k$ for each $k \in [m]$, it follows $v_k - v'_k = 0$ for each $k \in [m]$ by hypothesis. That is, $v_k = v'_k$ for all $k \in [m]$. Since v was arbitrary, this implies every vector in $V_1 + ... + V_m$ can only be written as a sum in one way,

so $V_1 + ... + V_m$ is a direct sum. Now, we prove the forward direction. Suppose $V_1 + ... + V_m$ is a direct sum. For each V_k, $0 \in V_k$ since every V_k is a subspace of V. Then, $0 = \sum_{k=1}^{m} 0 \in V_1 + ... + V_m$. Since $V_1 + ... + V_m$ is a direct sum, this is the only way to write 0 as a sum of vectors in V_k. □

Theorem 1.9 (direct sum of two subspaces)**.** *Suppose U and W are subspaces of V. Then, $U + W$ is a direct sum if and only if $U \cap W = \{0\}$.*

Proof. Suppose $U + W$ is a direct sum. Since $0 \in U$ and $0 \in W$, it follows that $0 \in U \cap W$. Suppose $v \in U \cap W$ and $v \neq 0$. Then, $v \in U$ and $v \in W$. So, $v = v + 0 \in U + W$ and $v = 0 + v \in U + W$. However, there can only be one way to write v as a sum of an element in U and an element in W since $U + W$ is a direct sum. Therefore, there does not exist $w \neq 0$ so that $w \in U \cap V$. Therefore, $U \cap W = \{0\}$. We prove the converse: suppose $U \cap W = \{0\}$. Suppose that there exists $u \in U$ and $w \in W$ so that $u + w = 0$. Then, $u = -w \in W$. So, $u \in U \cap W$. Therefore, $u = 0$ and $w = 0$. So, $0 = 0 + 0$ is the only way to write 0 in $U + W$. By Theorem 1.8, $U + W$ is a direct sum. □

2 Finite-Dimensional Vector Spaces

2.1 Span and Linear Independence

Note: When we refer to a **list** of vectors, we are always talking about a finite list.

Definition 2.1. A *linear combination* of a list $v_1, ..., v_m$ of vectors in V is a vector of the form:

$$\sum_{k=1}^{m} a_k v_k$$

where $a_k \in \mathbf{F}$ for all $k \in [m]$.

Definition 2.2. The set of all linear combinations of a list of vectors $v_1, ..., v_m$ in V is called the *span* of $v_1, ..., v_m$, denoted by $\text{span}(v_1, ..., v_m)$:

$$\text{span}(v_1, ..., v_m) = \left\{ \sum_{k=1}^{m} a_k v_k : a_k \in \mathbf{F} \text{ for all } k \in [m] \right\}$$

The span of the empty list is defined to be $\{0\}$. We say that $v_1, ..., v_n$ *spans* V if and only if $\text{span}(v_1, ..., v_n) = V$.

Theorem 2.1 (span is the smallest containing subspace)**.** *The span of a list of vectors in V is the smallest subspace of V containing all the vectors in the list.*

Proof. Let $v_1, ..., v_n$ be a list of vectors in V. We show that $\text{span}(v_1, ..., v_n)$ is a subspace of V. We can write $0 = \sum_{k=1}^{n} 0 \cdot v_k$, so $0 \in \text{span}(v_1, ..., v_n)$. Let $v \in \text{span}(v_1, ..., v_n)$. Then, $v = \sum_{k=1}^{n} c_k v_k$ where $c_k \in \mathbb{F}$ for all $k \in [n]$. Then, for every $\lambda \in \mathbb{F}$, $\lambda v = \sum_{k=1}^{n} (\lambda c_k) v_k \in \text{span}(v_1, ..., v_n)$. Now, let u and

w be vectors in $\text{span}(v_1, ..., v_n)$. Then, $u = \sum_{k=1}^{n} a_k v_k$ and $w = \sum_{k=1}^{n} b_k v_k$ where $a_k, b_k \in \mathbb{F}$ for all $k \in [n]$. Then, $u + w = \sum_{k=1}^{n} a_k v_k + \sum_{k=1}^{n} b_k v_k = \sum_{k=1}^{n} (a_k + b_k) v_k \in \text{span}(v_1, ..., v_n)$. Consequently, $u + w \in \text{span}(v_1, ..., v_n)$. So $\text{span}(v_1, ..., v_n)$ is a subspace of V. Suppose U is a subspace of V that contains v_k for all $k \in [n]$. Since U is a subspace, every arbitrary linear combination $\sum_{k=1}^{n} c_k v_k$ – where $c_k \in \mathbb{F}$ for all $k \in [n]$ – is contained in U. By Definition 2.2, $\text{span}(v_1, ..., v_n) \subset U$. That is, $\text{span}(v_1, ..., v_n)$ is the smallest subspace of V containing $v_1, ..., v_n$. □

Definition 2.3. A vector space is called *finite-dimensional* if some list – that is, a finite collection – of vectors in it spans the space. It is called *infinite-dimensional* if it is not finite-dimensional.

Definition 2.4. A function $p : \mathbb{F} \to \mathbb{F}$ is called a *polynomial* with coefficients in $\mathbb{F}$ if there exist $a_0, ..., a_m \in \mathbb{F}$ such that:

$$p(z) = a_0 + a_1 z + ... + a_m z^m$$

for all $z \in \mathbb{F}$. The set $\mathcal{P}(\mathbb{F})$ is the set of all polynomials with coefficients in $\mathbb{F}$. The polynomial $p \in \mathcal{P}(\mathbb{F})$ has *degree* m if there exists scalars $a_0, ..., a_m \in \mathbb{F}$ with $a_m \neq 0$ such that for every $z \in \mathbb{F}$, we have:

$$p(z) = a_0 + a_1 z + ... + a_m z^m$$

The polynomial that is identically 0 is said to have degree $-\infty$. The degree of a polynomial is denoted $\deg p$. The set of all polynomials with degree at most m with coefficients in $\mathbb{F}$ is denoted by $\mathcal{P}_m(\mathbb{F})$.

Remark 2.1. The set $P(\mathbb{F})$ is a vector space with addition and scalar multiplication defined as usual. It is a subspace of $\mathbb{F}^{\mathbb{F}}$. It is infinite-dimensional. The set $P_m(\mathbb{F})$ is also a vector space.

Definition 2.5. A list $v_1, ..., v_n$ of vectors in V is called *linearly independent* if the only choice of $a_1, ..., a_m \in \mathbb{F}$ that makes

$$\sum_{k=1}^{n} a_k v_k = 0$$

is $a_1 = ... = a_n = 0$. The empty list is defined to be linearly independent. If a list of vectors is not linearly independent, it is called *linearly dependent*.

Theorem 2.2 (linear independence and uniqueness)**.** *Let $v_1, ..., v_m$ be linearly independent. Then, every element of span$(v_1, ..., v_m)$ has a unique representation as a linear combination of $v_1, ..., v_m$.*

Proof. Let $v \in \text{span}(v_1, ..., v_m)$. Then, $v = \sum_{k=1}^{n} c_k v_k$ for some $c_k \in \mathbb{F}$ for all $k \in [n]$. Suppose there also exists coefficients $a_k \in \mathbb{F}$, $k \in [n]$, so that $v = \sum_{k=1}^{n} a_k v_k$. Then, $\sum_{k=1}^{n} (a_k - c_k) v_k = 0$. Since the $\{v_k\}_{k=1}^{n}$ are linearly independent, it follows that $a_k - c_k = 0$ for all $k \in [n]$. That is, $a_k = c_k$ for all $k \in [n]$. So, the representation of v is unique. □

Lemma 2.1 (linear dependence lemma)**.** *Suppose $v_1, ..., v_n$ is a linearly dependent list in V. Then, there exists $k \in [n]$ so that $v_k \in$ span$(v_1, ..., v_{k-1})$. Furthermore, if k satisfies the previous condition, then span$(v_1, ..., v_{k-1}, v_{k+1}, ..., v_n) =$ span$(v_1, ..., v_n)$.*

Proof. If $v_1, ..., v_k$ is linearly dependent, then if:

$$\sum_{k=1}^{n} c_k v_k = 0$$

there is some $k \in [n]$ so that $c_k \neq 0$. WLOG, suppose $c_n \neq 0$ (we can renumber the vectors and coefficients so that $c_n \neq 0$ if this is not true). Then, $\sum_{k=1}^{n-1} c_k v_k = -c_n v_n$. Therefore, $v_n = \sum_{k=1}^{n-1} \left(\frac{-c_k}{c_n}\right) v_k$. Therefore, $v_n \in \text{span}(v_1, ..., v_{n-1})$. For the next part, put $v_n = \sum_{k=1}^{n-1} a_k v_k$ for $a_k \in \mathbb{F}$, where $k \in [n]$. Then, if $v = \sum_{k=1}^{n} c_k v_k$:

$$\begin{aligned} v &= \sum_{k=1}^{n-1} c_k v_k + c_n v_n \\ &= \sum_{k=1}^{n-1} c_k v_k + \sum_{k=1}^{n-1} c_n a_k v_k \\ &= \sum_{k=1}^{n-1} (c_k + c_n a_k) v_k \end{aligned}$$

We have shown if $v \in \text{span}(v_1, ..., v_n)$, then $v \in \text{span}(v_1, ..., v_{n-1})$, implying that $\text{span}(v_1, ..., v_n) \subset \text{span}(v_1, ..., v_{n-1})$. Now, suppose that $v = \sum_{k=1}^{n-1} c_k v_k$. Put $c_n = 0$. Then, $v = \sum_{k=1}^{n} c_k v_k$. So, we have shown that if $v \in \text{span}(v_1, ..., v_{n-1})$, then $v \in \text{span}(v_1, ..., v_n)$. That is, $\text{span}(v_1, ..., v_{n-1}) \subset \text{span}(v_1, ..., v_n)$. So, $\text{span}(v_1, ..., v_n) = \text{span}(v_1, ..., v_{n-1})$. □

Theorem 2.3 (length of linearly independent list $\leq$ length of spanning list)**.** *In a finite-dimensional vector space, the length of every linearly independent list of vectors is less than or equal to the length of every spanning list of vectors.*

Proof. Let V be a finite-dimensional vector space so that $V = \text{span}(v_1, ..., v_n)$. Let $w_1, ..., w_m \in V$ be a linearly independent list. We show $m \leq n$. Since $w_1 \in \text{span}(v_1, ..., v_n)$, the list $w_1, v_1, ..., v_n$ is a linearly dependent list. By Lemma 2.1, one of the vectors in the list $w_1, v_1, ..., v_n$ can be written as a linear combination of the previous vectors. It cannot be that w_1 is a linear combination of the "previous" vectors in the list because the span of the empty list is $\{0\}$, and $w_1 \neq 0$ since $w_1, ..., w_m$ is a linearly independent list. So, we can remove one of the v_k from the list to obtain a new list (which we will call B) that is of length n which contains w_1 and the remaining v_k's; B still spans V. Now, suppose we continue this process and arrive at step j, $1 < j \leq m$. Then, the list B from step

$j-1$ is of length n and spans V. In particular, w_j is in the span of B. If we insert w_j after $w_1, ..., w_{j-1}$ into B, then the new list has length $n+1$ and is linearly dependent. Therefore, by Lemma 2.1, there exists a vector in this list that can be written as the linear combination of the previous vectors. We know that w_j cannot be this vector since $w_1, ..., w_j$ are linearly independent. Therefore, it must be one of the v_k's. Once we delete one of the v_k's, the remaining list is of length n and now consists of $w_1, ..., w_j$ and the remaining v_k's; this list spans V. This process will terminate since the list of v_k's is finite. At the end of each step, the list is always of length n, so there must be at most as many w_j's as v_k's, i.e, $m \leq n$. □

Theorem 2.4 (every subspace of finite-dimensional space is finite-dimensional)**.** *Every subspace of a finite-dimensional vector space is finite-dimensional.*

Proof. Let V be a finite-dimensional vector space. Then, let U be a subspace of V. Choose $u_1 \in U$ arbitrarily. Then, if $U = \text{span}(u_1)$, we are done. We generalize this process for $k > 1$: if $U = \text{span}(u_1, ..., u_{k-1})$, then U is finite-dimensional and we are done. Otherwise, choose $u_k \notin \text{span}(u_1, ..., u_{k-1})$. By Lemma 2.1, the list is linearly independent. This process will terminate since each $u_k \in V$ and it cannot exceed the spanning list of V by Theorem 2.3. Therefore, U is a finite-dimensional subspace. □

2.2 Bases

Definition 2.6. A *basis* of V is a list of vectors in V that is linearly independent and spans V.

Theorem 2.5 (criterion for basis)**.** *A list $v_1, ..., v_n$ of vectors in V is a basis of V if and only if every $v \in V$ can be written uniquely in the form:*

$$v = \sum_{k=1}^{n} a_k v_k$$

where $a_1, ..., a_n \in \mathbb{F}$

Proof. Suppose $v_1, ..., v_n$ is a basis for V. By Theorem 2.2, every element of $\text{span}(v_1, ..., v_n)$ can be written uniquely as a linear combination of $v_1, ..., v_n$. Since $\text{span}(v_1, ..., v_n) = V$, every element of V can be written uniquely as a linear combination of $v_1, ..., v_n$. Conversely, suppose every element of V can be written uniquely as a linear combination of $v_1, ..., v_n$. Then, since every element of V can be written as a linear combination of $v_1, ..., v_n$, $V = \text{span}(v_1, ..., v_n)$. Suppose for some coefficients $c_k \in \mathbb{F}$:

$$\sum_{k=1}^{n} c_k v_k = 0$$

Then, since $0 \in V$ and $0 = \sum_{k=1}^{n} 0 \cdot v_k$, it follows that $c_k = 0$ for each $k \in [n]$. Therefore, $v_1, ..., v_n$ form a linearly independent set in V. By Definition 2.6, $v_1, ..., v_n$ form a basis for V. □

Theorem 2.6 (every spanning list contains a basis)**.** *Every spanning list in a vector space can be reduced to a basis of the vector space.*

Proof. Let $B_0 = (v_1, ..., v_n)$ be a list and V be a vector space so that $V = \text{span } B_0$. If B_0 is linearly independent, then we are done. Otherwise, by Lemma 2.1, there exists an element in the list B_0 that can be removed to obtain a list B_1 so that $V = \text{span } B_1$. We can again invoke Lemma 2.1 to obtain a list B_2 so that $V = \text{span } B_2$ if B_1 is not linearly independent. The process will eventually terminate since lists are finite and the resulting list will span the space and also be linearly independent and hence a basis by Definition 2.6. □

Corollary 2.1 (every finite-dimensional vector space has a basis)**.** *Every finite-dimensional vector space has a basis.*

Proof. Let V be a finite-dimensional vector space. Then, $V = \text{span}(v_1, ..., v_n)$ for some vectors $v_1, ..., v_n \in V$. By Theorem 2.6, this list can be reduced to a basis of V. □

Theorem 2.7 (every linearly independent list of vectors can be extended to basis of vector space)**.** *Every linearly independent list of vectors in a finite-dimensional vector space can be extended to a basis of the vector space.*

Proof. Let $v_1, ..., v_m$ be a linearly independent set in a finite-dimensional vector space V. Let $v'_1, ..., v'_n$ be a list of vectors that spans V. Then, the list $v_1, ..., v_m, v'_1, ..., v'_n$ span V. By Theorem 2.6, we can reduce this list to a basis of V. None of the $v_1, ..., v_m$ are deleted from the list since they are a linearly independent set: the procedure used in Theorem 2.6 only removes vectors that are in the span of the vectors preceding it. □

Theorem 2.8 (every subspace of V is part of a direct sum equal to V)**.** *Suppose V is finite-dimensional and U is a subspace of V. Then there is a subspace W of V such that $V = U \oplus W$.*

Remark 2.2. This result also holds for infinite-dimensional spaces.

Proof. By Theorem 2.4, U is a finite-dimensional subspace. By Corollary 2.1, there exists a basis $u_1, ..., u_n$ for U. Then, $u_1, ..., u_n$ is a linearly independent list in V. By Theorem 2.7, we can extend this list to a basis $u_1, ..., u_n, w_1, ..., w_m$ of V. Let $W = \text{span}(w_1, ..., w_n)$. We want to show that $V = U \oplus W$. This amounts to showing that $V = U + W$ and that $U \cap W = \{\emptyset\}$ by Theorem 1.9. Let $v \in V$. Then, because $u_1, ..., u_n, w_1, ..., w_m$ spans V, it follows that $v = \sum_{k=1}^{n} a_k u_k + \sum_{j=1}^{m} b_j w_j$ where $a_k \in \mathbb{F}$ and $b_j \in \mathbb{F}$ for all $k \in [n]$ and for all $j \in [m]$. Then, $\sum_{k=1}^{n} a_k u_k \in U$ and $\sum_{j=1}^{m} b_j w_j \in W$. So, $v \in U + W$. It follows that $V = U + W$. Suppose that $v \in U \cap W$. Then, $v = \sum_{k=1}^{n} a_k u_k$ and $v = \sum_{j=1}^{m} b_j w_j$ for coefficients in $a_k \in \mathbb{F}$ and $b_j \in \mathbb{F}$. Then, $\sum_{k=1}^{n} a_k u_k =$

$\sum_{j=1}^{m} b_j w_j$, so $\sum_{k=1}^{n} a_k u_k + \sum_{j=1}^{m}(-b_j)w_j = 0$. Since $u_1, ..., u_n, w_1, ..., w_m$ is a linearly independent list in V, it follows that $a_k = 0$ and $b_j = 0$ for all $k \in [n]$ and $j \in [m]$. Therefore, $v = 0$, so $U \cap W = \{0\}$. □

2.3 Dimension

Theorem 2.9 (basis length does not depend on basis)**.** *Any two bases of a finite-dimensional vector space have the same length.*

Proof. Let B_1 and B_2 be two bases for a finite-dimensional vector space V (we know that V has at least one basis by Corollary 2.1). Then, by Theorem 2.3, since B_1 is a linearly independent list and B_2 spans V, the length $B_1 \leq$ length B_2. Again, by Theorem 2.3, since B_2 is a linearly independent list and B_1 spans V, length $B_2 \leq$ length B_2. Consequently, length $B_1 =$ length B_2. □

So, the length of the basis is the same, meaning we can consistently define the dimension of a vector space as follows:

Definition 2.7. The *dimension* of a finite-dimensional vector space V is the length of any basis of the vector space. It is denoted $\dim V$.

Theorem 2.10 (dimension of a subspace)**.** *If V is a finite-dimensional vector space and U is a subspace of V, then* $\dim U \leq \dim V$.

Proof. Since U is a subspace of V and V is finite-dimensional, U is finite-dimensional by Theorem 2.4. Let B_U be a list of vectors that is a basis for U. Let B_V be a list of vectors that is a basis for V. Since B_U is a linearly independent set of vectors in V and B_V is a list that spans V, $\dim U =$ length $B_u \leq$ length $B_V = \dim V$ by Theorem 2.3. □

Theorem 2.11 (linearly independent list of the right length is a basis)**.** *Suppose V is finite-dimensional. Then every linearly independent list of vectors in V of length* $\dim V$ *is a basis of V.*

Proof. Put $n = \dim V$. Let $u_1, ..., u_n$ be a linearly independent list in V. Since V is finite-dimensional, it has a basis by Corollary 2.1. Moreover, this basis is of length n by Theorem 2.9. By Theorem 2.7, we can extend $u_1, ..., u_n$ to be a basis of V. Let $u_1, ..., u_n, v_1, ..., v_m$ be this extended list which is a basis for V. Then, $\dim V = n + m$ by Theorem 2.9. But $\dim V = n \neq n + m$ by hypothesis. So, $u_1, ..., u_n$ is already a basis for V. □

Theorem 2.12 (subspace of full dimension equals whole space)**.** *Suppose that V is finite-dimensional and U is a subspace of V such that* $\dim U = \dim V$. *Then $U = V$.*

Proof. Since V is finite-dimensional, it has a basis by Corollary 2.1. Since U is a subspace of V and V is finite-dimensional, U has a basis by Theorem 2.4. Put $n = \dim V = \dim U$. Then, by Theorem 2.9, the basis $u_1, ..., u_n$ of U has length n. Since $u_1, ..., u_n$ is a linearly independent list in V of size $n = \dim U = \dim V$,

$u_1, ..., u_n$ is a basis for V by Theorem 2.11. Then, every vector in V is in $\text{span}(u_1, ..., u_n) = U$, so $U = V$. □

Theorem 2.13 (spanning list of right length is a basis)**.** *Suppose V is finite-dimensional. Then every spanning list of vectors in V of length $\dim V$ is a basis of V.*

Proof. Since V is finite-dimensional, it has a basis by Corollary 2.1. Let $v_1, ..., v_n$ be a spanning list in V where $n = \dim V$. By Theorem 2.6, this list contains a basis for V. Suppose we retain the vectors $v_1, ..., v_m$ where $m < n$. Then, $v_1, ..., v_m$ is a basis for V. Then, $\dim V = m$ by Theorem 2.9. However, this implies $m \neq n$, which contradicts the hypothesis. Therefore, $v_1, ..., v_n$ is already a basis for V. □

Theorem 2.14 (dimension of a sum)**.** *If V_1 and V_2 are subspaces of a finite-dimensional vector space, then* $\dim(V_1 + V_2) = \dim V_1 + \dim V_2 - \dim(V_1 \cap V_2)$.

Proof. Let $v_1, ..., v_m$ be a basis of $V_1 \cap V_2$. Then, $\dim(V_1 \cap V_2) = m$. By Theorem 2.7, we can extend the list to a basis of V_1 and obtain $v_1, ..., v_m, u_1, ..., u_n$. Therefore, $\dim V_1 = m+n$. Similarly, by Theorem 2.7, we can extend the basis of $V_1 \cap V_2$ to a basis of V_2 and obtain $v_1, ..., v_m, w_1, ..., w_l$. Therefore $\dim V_2 = m+l$. Then, we show that $v_1, ..., v_m, u_1, ..., u_n, w_1, ..., w_l$ is a basis of $V_1 + V_2$ since then $\dim(V_1+V_2) = m+n+l = (m+n)+(m+l)-m = \dim V_1+\dim V_2-\dim(V_1 \cap V_2)$. First, $v_1, ..., v_m, u_1, .., u_n, w_1, ..., w_l$ is contained in $V_1 \cup V_2$, so it is in the subspace $V_1 + V_2$ by Theorem 1.7. Since the list consists solely of vectors in the bases of V_1 and V_2, $\text{span}(v_1, ..., v_m, u_1, ..., u_n, w_1, ..., w_l) = V_1 + V_2$. So, we only need to show that $v_1, ..., v_m, u_1, ..., u_n, w_1, ..., w_l$ is linearly independent. Suppose that for some arbitrary coefficients that:

$$\sum_{k=1}^{m} a_k v_k + \sum_{k=1}^{n} b_k u_k + \sum_{k=1}^{l} c_k w_k = 0$$

Then, $\sum_{k=1}^{l} c_k w_k = -\sum_{k=1}^{m} a_k v_k - \sum_{k=1}^{n} b_k u_k$. This implies that $\sum_{k=1}^{l} c_k w_k \in V_1$. Therefore, $\sum_{k=1}^{l} c_k w_k = \sum_{k=1}^{m} d_k v_k$ where $d_k \in \mathbb{F}$ for each $k \in [m]$. However, $v_1, ..., v_m, w_1, ..., w_l$ is linearly independent, so $c_k = 0$ for all $k \in [l]$ and $d_k = 0$ for all $k \in [m]$. So, we get that $\sum_{k=1}^{m} a_k v_k + \sum_{k=1}^{n} b_k u_k = 0$. Since $v_1, ..., v_m, u_1, ..., u_n$ is linearly independent, $a_k = 0$ for all $k \in [m]$ and $b_k = 0$ for all $k \in [n]$. Therefore, $v_1, ..., v_m, u_1, ..., u_n, w_1, ..., w_l$ is a basis for $V_1 + V_2$. □

Corollary 2.2 (dimension of direct sum)**.** *If V_1 and V_2 are subspaces of a finite-dimensional vector space and $V_1 + V_2$ is a direct sum, then* $\dim(V_1 \oplus V_2) = \dim V_1 + \dim V_2$.

Proof. If $V_1 + V_2$ is a direct sum, then $V_1 \cap V_2 = \{0\}$ by Theorem 1.9. Then, $\dim\{0\} = 0$, so $\dim(V_1 \oplus V_2) = \dim V_1 + \dim V_2$ by Theorem 2.14. □

3 Linear Maps

3.1 Definitions

Definition 3.1. A *linear map* from V to W is a function $T : V \to W$ with the following properties:

1. **additivity**: $T(u + v) = T(u) + T(v)$ for all $u, v \in V$
2. **homogeneity**: $T(\lambda v) = \lambda T(V)$ for all $\lambda \in \mathbb{F}$ and $v \in V$

We will write Tv to denote $T(v)$. The set of linear maps from V to W is denoted by $\mathcal{L}(V, W)$. We will write $\mathcal{L}(V) = \mathcal{L}(V, V)$. We use 0 to denote the linear map $0v = 0$ for all $v \in V$ ($0 \in \mathcal{L}(V, W)$). We will use $I \in \mathcal{L}(V)$ to denote the *identity operator*, where $Iv = v$ for all $v \in V$.

Lemma 3.1 (linear map lemma)**.** *Suppose $v_1, ..., v_n$ is a basis of V and $w_1, ..., w_n \in W$. Then there exists a unique linear map $T : V \to W$ such that $Tv_k = w_k$ for each $k \in [n]$.*

Proof. Let us define a map $T : V \to W$ so that $Tv_k = w_k$ for each $k \in [n]$. Then, for each $v \in V$, define $Tv = \sum_{k=1}^{n} c_k w_k$ where each of the $c_k \in \mathbb{F}$ and $v = \sum_{k=1}^{n} c_k v_k$. Since v_k is a basis for V, the coefficients c_k are unique by Theorem 2.5, so T is well-defined. We prove that T is linear. Let $u, v \in V$. Then, $u = \sum_{k=1}^{n} a_k v_k$ and $v = \sum_{k=1}^{n} b_k v_k$ where $a_k, b_k \in \mathbb{F}$ for each $k \in [n]$. It is easy to check that $T(u + v) = T(u) + T(v)$. Next, we show $T(\lambda v) = \lambda T(v)$ for each $v \in V$ and $\lambda \in \mathbb{F}$. Since $v \in V$, $v = \sum_{k=1}^{n} c_k v_k$ where $c_k \in \mathbb{F}$ for each $k \in [n]$. Then, $T(\lambda v) = T(\sum_{k=1}^{n} (c_k \lambda) v_k) = \sum_{k=1}^{n} (\lambda c_k) w_k = \lambda \sum_{k=1}^{n} c_k w_k = \lambda T(v)$. So, T is a linear map. Suppose there exists another linear map $T' : V \to W$ so that $T'(v_k) = w_k$ for each $k \in [n]$. Let $v \in V$. Then, $v = \sum_{k=1}^{n} c_k v_k$ where $c_k \in \mathbb{F}$ for each $k \in [n]$. Then, $T(v) = \sum_{k=1}^{n} c_k w_k$. Since T' is linear:

$$\begin{aligned} T'(v) &= T'\Big(\sum_{k=1}^{n} c_k v_k\Big) \\ &= \sum_{k=1}^{n} c_k T'(v_k) && \text{Definition 3.1} \\ &= \sum_{k=1}^{n} c_k w_k \\ &= T(v) \end{aligned}$$

So, $T \equiv T'$, meaning T is unique. □

Definition 3.2 (addition and scalar multiplication on $\mathcal{L}(V, W)$)**.** Suppose $S, T \in \mathcal{L}(V, W)$ and $\lambda \in \mathbb{F}$. The *sum* $S + T$ and the product λT are linear maps from

V to W defined by:

$$(S+T)(v) = Sv + Tv$$
$$(\lambda T)(v) = \lambda(Tv)$$

for all $v \in V$.

Theorem 3.1. *With the operation of addition and scalar multiplication defined in Definition 3.2, $\mathcal{L}(V,W)$ is a vector space.*

Proof. Let $S, T, R \in \mathcal{L}(V,W)$, $u, v \in V$, and $\lambda \in \mathbb{F}$.

(a) Commutativity: Then, $(S+T)v = Sv + Tv$. $(T+S)v = Tv + Sv$. Since $Sv, Tv \in W$, $Sv + Tv = Tv + Sv$. Therefore, $S + T \equiv T + S$.

(b) Associativity: Then, $((S+T)+R)v = (S+T)v + Rv = Sv + Tv + Rv = Sv + (Tv + Rv) = Sv + (T+R)v = (S + (T+R))v$.

(c) Additive Identity: $(S+0)v = Sv+0v = Sv+0 = Sv$. Therefore, $S+0 \equiv S$.

(d) Additive Inverse: Since S is linear, $-S := -1 \cdot S$ is linear. So, $(S + (-S))v = Sv + (-S)v = Sv + -(Sv) = 0$. Since this is true for all $v \in V$, $S + (-S) \equiv 0$.

(e) Multiplicative identity: By how multiplication is defined, $1 \cdot T = T$ for all $T \in \mathcal{L}(V,W)$.

(f) Distributive Property: $(\lambda(S+T))v = \lambda \cdot ((S+T)v) = \lambda \cdot (Sv + Tv) = \lambda(Sv) + \lambda(Tv) = (\lambda S)(v) + (\lambda T)v$. So, $\lambda(S+T) \equiv \lambda S + \lambda T$. Let $a, b \in \mathbb{F}$. Then, $((a+b)S)v = (a+b) \cdot (Sv) = a \cdot Sv + b \cdot Sv = (aS)v + (bS)v$. Therefore, $(a+b)S \equiv aS + bS$.

By Definition 1.2, it is a vector space. □

Definition 3.3 (product of linear maps)**.** If $T \in \mathcal{L}(V,W)$ and $S \in \mathcal{L}(V,W)$, then the *product* $ST \in \mathcal{L}(V,W)$ is defined by $(ST)u = S(Tu)$ for all $u \in U$.

Theorem 3.2 (algebraic properties of products of linear maps)**.**

1. ***associativity**: $(T_1T_2)T_3 = T_1(T_2T_3)$ whenever the domains are defined appropriately for linear maps T_1, T_2, T_3.*
2. ***identity**: $TI = IT = T$ whenever $T \in \mathcal{L}(V,W)$.*
3. ***distributive properties**: $(S_1 + S_2)T = S_1T + S_2T$ and $S(T_1 + T_2) = ST_1 + ST_2$ whenever $T, T_1, T_2 \in \mathcal{L}(V,W)$ and $S, S_1, S_2 \in \mathcal{L}(V,W)$.*

Proof.

1. **associativity**: Let $v \in V$. Let T_1, T_2, T_3 be linear maps where the domains are defined appropriately. Then, $((T_1T_2)T_3)v = (T_1T_2)(T_3v) = T_1(T_2(T_3v)) = T_1((T_2T_3)v)$.

2. **identity**: Let $v \in V$ and $T \in \mathcal{L}(V, W)$. Then $(TI)v = T(Iv) = T(v) = Tv$. And, $(IT)v = I(Tv) = Tv$. Therefore, $(TI)v = (IT)v = Tv$, so $TI = IT = T$. Note that the identity operator in TI is in $\mathcal{L}(V)$ whereas the identity operator in IT is in $\mathcal{L}(W)$.

3. **distributive properties**: Let $v \in V$. Then, $((S_1 + S_2)T)v = (S_1 + S_2)(Tv) = S_1(Tv) + S_2(Tv) = (S_1T)v + (S_2T)v = (S_1T + S_2T)v$. And $(S(T_1 + T_2))v = S((T_1 + T_2)v) = S(T_1v + T_2v) = S(T_1v) + S(T_2v) = (ST_1)v + (ST_2)v$.

□

Theorem 3.3 (linear maps take 0 to 0)**.** *Suppose $T : V \to W$ is a linear map. Then $T(0) = 0$.*

Proof. Let $v \in V$ be arbitrary. Then, $Tv = T(v) = T(v + 0)$. By linearity, $T(v + 0) = T(v) + T(0)$. Therefore, $T(0) = 0$. □

3.2 Null Spaces and Ranges

Definition 3.4 (null space)**.** For $T \in \mathcal{L}(V, W)$, the *null space* of T, denoted by $\operatorname{null} T$, is the subset of V consisting of those vectors that T maps to 0:

$$\operatorname{null} T = \{v \in V : Tv = 0\}$$

Theorem 3.4 (null space is a subspace)**.** *Suppose $T \in \mathcal{L}(V, W)$. Then, $\operatorname{null} T$ is a subspace of V.*

Proof. We show this using Theorem 1.6. First, suppose that $v, w \in \operatorname{null} T$. Then, $Tv = 0$ and $Tw = 0$, so $T(v + w) = 0$ since T is linear. Therefore, $v + w \in nullspaceT$. Next, let $\lambda \in \mathbb{F}$ and $v \in \operatorname{null} T$. Then, by the linearity of T:

$$\begin{aligned} Tv &= 0 \\ \lambda(Tv) &= 0 \\ T(\lambda v) &= 0 \qquad \text{linearity of } T \end{aligned}$$

Therefore, $\lambda v \in \operatorname{null} T$. Finally, by Theorem 3.3, $T(0) = 0$, so $0 \in \operatorname{null} T$. □

Theorem 3.5 (injectivity iff null space equals 0)**.** *Let $T \in \mathcal{L}(V, W)$. Then, T is injective if and only if $\operatorname{null} T = \{0\}$.*

Proof. Suppose T is injective. Let $v \in V$ so that $Tv = 0$. Since T is linear, Theorem 3.3 implies that $T(0) = 0$. Therefore, $T(v) = T(0)$. By injectivity, $v = 0$. Consequently, $\operatorname{null} T = \{0\}$. Suppose now that $\operatorname{null} T = \{0\}$. Suppose that $Tu = Tv$. Then, the linearity of T implies that $T(u - v) = 0$. Therefore, $u - v \in \operatorname{null} T$, so $u - v = 0$. That is, $u = v$. Since u and v were arbitrary, T is injective. □

Definition 3.5 (range)**.** For $T \in \mathcal{L}(V, W)$, the *range* of T is the subset of W consisting of those vectors that are equal to Tv for some $v \in V$:

$$\operatorname{range} v = \{Tv : v \in V\}$$

Theorem 3.6 (the range is a subspace)**.** *If $T \in \mathcal{L}(V, W)$, then* $\operatorname{range} T$ *is a subspace of W.*

Proof. We prove this via Theorem 1.6. First, let $u, w \in \operatorname{range} T$. Then, $u = Tv_1$ and $w = Tv_2$ for some $v_1, v_2 \in V$. So, $u + w = Tv_1 + Tv_2 = T(v_1 + v_2)$ since T is linear. Then, $u + w \in \operatorname{range} T$ since $v_1 + v_2 \in V$. Next, let $\lambda \in \mathbb{F}$ and $u \in \operatorname{range} T$. Then, $u = Tv$ for some $v \in V$. By the linearity of T, $\lambda u = \lambda(Tv) = T(\lambda v)$. Since $\lambda v \in V$, $\lambda u \in \operatorname{range} T$. If we take $\lambda = 0$, then $0 \in \operatorname{range} T$. □

Theorem 3.7 (The Fundamental Theorem of Linear Maps)**.** *Suppose V is finite-dimensional and $T \in \mathcal{L}(V, W)$. Then,* $\operatorname{range} T$ *is finite-dimensional and* $\dim V = \dim \operatorname{null} T + \dim \operatorname{range} T$*.*

Proof. Since V is finite-dimensional, it has a basis, $v_1, ..., v_n$. Then, for any $w \in \operatorname{range} T$:

$$\begin{aligned} w &= Tv \\ &= T\Big(\sum_{k=1}^{n} c_k v_k\Big) && \text{for } c_k \in \mathbb{F} \\ &= \sum_{k=1}^{n} c_k (Tv_k) && \text{linearity of } T \end{aligned}$$

So, the vectors $Tv_1, ..., Tv_n$ span $\operatorname{range} T$. By Theorem 2.6, this spanning list can be reduced to a basis of $\operatorname{range} T$. Hence, $\operatorname{range} T$ is finite-dimensional. Now, $\operatorname{null} T$ is a subspace of V by Theorem 3.4. Therefore, it is finite-dimensional by Theorem 2.4, so it has a basis $u_1, ..., u_n$. Since this is a linearly independent list, we can extend it to a basis for V and obtain $u_1, ..., u_n, v_1, ..., v_m$ (Theorem 2.7). We show that the vectors $Tv_1, ..., Tv_m$ are a basis for $\operatorname{range} T$. First, we show that they span $\operatorname{range} T$. Let $w \in \operatorname{range} T$. Then:

$$\begin{aligned} w &= Tv \\ &= T\Big(\sum_{k=1}^{n} c_k u_k + \sum_{k=1}^{m} d_k v_k\Big) \\ &= \Big(\sum_{k=1}^{n} c_k (Tu_k) + \sum_{k=1}^{m} d_k (Tv_k)\Big) && \text{By linearity of } T \\ &= \sum_{k=1}^{n} c_k \cdot 0 + \sum_{k=1}^{m} d_k (Tv_k) && u_k \in \operatorname{null} T \\ &= \sum_{k=1}^{m} d_k (Tv_k) \end{aligned}$$

Therefore, $w \in \text{span}(v_1, ..., v_m)$. Now, we show that $Tv_1, ..., Tv_m$ are linearly independent. Suppose that:

$$\sum_{k=1}^{n} c_k(Tv_k) = 0$$

Then:

$$T\Big(\sum_{k=1}^{n} c_k v_k\Big) = 0$$

So, $\sum_{k=1}^{n} c_k v_k \in \text{null}\, T$. Therefore, it is in the span of $u_1, ..., u_n$, and:

$$\sum_{k=1}^{m} c_k v_k = \sum_{k=1}^{n} d_k u_k$$

$$\sum_{k=1}^{m} c_k v_k - \sum_{k=1}^{n} d_k u_k = 0$$

Since $v_1, ..., v_m, u_1, ..., u_n$ are linearly independent, it follows that $d_k = 0$ for all $k \in [n]$ and $c_k = 0$ for all $k \in [m]$. So, $Tv_1, ..., Tv_m$ are linearly independent. Therefore, it is a basis for range T. Since $\dim \text{null}\, T = n$ and $\dim \text{range}\, T = m$ and $u_1, ..., u_n, v_1, ..., v_m$ are a basis for V, it follows that $\dim V = m + n$, or:

$$\dim V = \dim \text{null}\, T + \dim \text{range}\, T$$

□

Theorem 3.8 (linear map to a lower dimensional space is not injective)**.** *Suppose V and W are finite-dimensional vector spaces such that* $\dim V > \dim W$*. Then no linear map from V to W is injective.*

Proof. Suppose for contradiction that $T : V \to W$ is a linear map that is injective. Then, $\text{null}\, T = \{0\}$ by Theorem 3.5. By Theorem 3.6 and *Theorem* 2.10, $\dim \text{range}\, T \leq \dim W$. By Theorem 3.7:

$$\begin{aligned} \dim V &= \dim \text{null}\, T + \dim \text{range}\, T \\ &= 0 + \dim \text{range}\, T \\ &\leq \dim W \end{aligned}$$

But this contradicts the hypothesis that $\dim V > \dim W$. Therefore, there is no linear map from V to W that is injective if $\dim V > \dim W$. □

Theorem 3.9 (linear map to a higher-dimensional subspace is not surjective)**.** *Suppose V and W are finite-dimensional vector spaces such that* $\dim V < \dim W$*. Then no linear map from V to W is surjective.*

Proof. Suppose for contradiction that $T : V \to W$ is a linear map which is surjective. Then, $\dim \operatorname{range} T = \dim W$. Therefore, by Theorem 3.7, $\dim V = \dim \operatorname{null} T + \dim W$. Note that $\dim \operatorname{null} T \geq 0$, so $\dim V \geq \dim W$, contradicting the initial hypothesis. Therefore, there is no linear map from V to W that is surjective if $\dim V < \dim W$. □

Definition 3.6. Fix positive integers m, n and let $A_{j,k} \in \mathbb{F}$ for $j \in [m]$ and $k \in [n]$ and arbitrary $c_1, ..., c_m \in \mathbb{F}$:

$$\sum_{k=1}^{n} A_{1,k} x_k = c_1$$

$$\vdots$$

$$\sum_{k=1}^{n} A_{m,k} x_k = c_m$$

This is a *homogeneous* system of linear equations whenever $c_1 = ... = c_m = 0$. It is *inhomogeneous* if at least one of the $c_k \neq 0$.

We consider whether there exist nonzero solutions for the homogeneous system of linear equations. Define $T : \mathbb{F}^n \to \mathbb{F}^m$ by $T(x_1, ..., x_n) = (\sum_{k=1}^{n} A_{1,k} x_k, ..., \sum_{k=1}^{n} A_{m,k} x_k)$. This is a linear map, and we are interested in determining where $T(x_1, ..., x_n) = 0$ (the zero vector, not the number in this case). We know that $x_1 = ... = x_n = 0$ is a solution. The question is if there are more solutions. That is, is $\operatorname{null} T \neq \{0\}$? By Theorem 3.5, we are equivalently asking whether T is not injective.

Theorem 3.10 (homogeneous system of linear equations)**.** *A homogeneous system of linear equations with more variables than equations has nonzero solutions.*

Proof. By Theorem 3.5, we only need to show that $T : \mathbb{F}^n \to \mathbb{F}^m$ is not injective. Since $n > m$, Theorem 3.8 implies that T is not injective since it is a linear map to a lower-dimensional space. □

We now consider an inhomogeneous system of linear equations and prove an analogous result. Define $T : \mathbb{F}^n \to \mathbb{F}^m$ by $T(x_1, ..., x_n) = (\sum_{k=1}^{n} A_{1,k} x_k, ..., \sum_{k=1}^{n} A_{m,k} x_k)$. This is a linear map and we are interested when $T(x_1, ..., x_n) = (c_1, ..., c_m)$ has no solutions for a particular choice of $c_1, ..., c_m$. That is, we are interested in knowing when $\operatorname{range} T \neq \mathbb{F}^m$.

Theorem 3.11 (inhomogeneous system of linear equations)**.** *An inhomogeneous system of linear equations with more equations than variables has no solutions for some choice of constant terms.*

Proof. Using the notation from above, the number of equations is m and the number of variables is n. We are given that $m > n$. Then $T : \mathbb{F}^n \to \mathbb{F}^m$ is a linear map from a lower dimensional space to a higher dimensional space. By Theorem 3.9, T cannot be surjective. Therefore, $\operatorname{range} T \neq \mathbb{F}^m$. □

3.3 Matrices

Definition 3.7. Suppose m and n are nonnegative integers. An m-by-n matrix A is a rectangular array of elements of $\mathbb{F}$ with m rows and n columns:

$$\begin{bmatrix} A_{1,1} & ... & A_{1,n} \\ \vdots & \ddots & \vdots \\ A_{m,1} & ... & A_{m,n} \end{bmatrix}$$

The notation $A_{j,k}$ denotes the entry in row j, column k.

Definition 3.8. Suppose $T \in \mathcal{L}(V,W)$ and $v_1, ..., v_n$ is a basis of V and $w_1, ..., w_m$ is a basis of W. The *matrix* of T with respect to these bases is the m-by-n matrix $\mathcal{M}(T)$ whose entries $A_{j,k}$ are defined by:

$$Tv_k = \sum_{j=1}^{m} A_{j,k} w_j$$

If the bases $v_1, ..., v_n$ and $w_1, ..., w_m$ are not clear from the context, then the notation $\mathcal{M}(T, (v_1, ..., v_n), (w_1, ..., w_m))$ is used.

Remark 3.1. To represent each Tv_k, we use the scalars in the kth column of the matrix $\mathcal{T}$ and then do a "dot product" with the basis vectors $w_1, ..., w_m$.

Now, we define a scalar multiplication and addition for matrices of linear maps:

Definition 3.9. The *sum of two matrices of the same size* is obtained by adding the corresponding entries of the matrix:

$$\begin{bmatrix} A_{1,1} & ... & A_{1,n} \\ \vdots & \ddots & \vdots \\ A_{m,1} & ... & A_{m,n} \end{bmatrix} + \begin{bmatrix} C_{1,1} & ... & C_{1,n} \\ \vdots & \ddots & \vdots \\ C_{m,1} & ... & C_{m,n} \end{bmatrix} = \begin{bmatrix} A_{1,1} + C_{1,1} & ... & A_{1,n} + C_{1,n} \\ \vdots & \ddots & \vdots \\ A_{m,1} + C_{m,1} & ... & A_{m,n} + C_{m,n} \end{bmatrix}$$

The *scalar multiple of a matrix* is defined as follows:

$$\lambda \begin{bmatrix} A_{1,1} & ... & A_{1,n} \\ \vdots & \ddots & \vdots \\ A_{m,1} & ... & A_{m,n} \end{bmatrix} = \begin{bmatrix} \lambda A_{1,1} & ... & \lambda A_{1,n} \\ \vdots & \ddots & \vdots \\ \lambda A_{m,1} & ... & \lambda A_{m,n} \end{bmatrix}$$

Theorem 3.12 (matrix of sum of linear maps)**.** *Suppose $S, T \in \mathcal{L}(V,W)$. Then, $\mathcal{M}(S+T) = \mathcal{M}(S) + \mathcal{M}(T)$.*

Proof. Let $v_1, ..., v_n$ be a basis for V and $w_1, ..., w_m$ be a basis for W. Let $A = \mathcal{M}(S)$ and $B = \mathcal{M}(T)$. Then:

$$Sv_k = \sum_{j=1}^{m} A_{j,k} w_j$$

$$Tv_k = \sum_{j=1}^{m} B_{j,k} w_j$$

So, $(S+T)v_k = \sum_{j=1}^{m}(A_{j,k}+B_{j,k})w_j$. Therefore, $\mathcal{M}(S+T) = A+B = \mathcal{M}(S)+\mathcal{M}(T)$. □

Theorem 3.13 (matrix of scalar multiple of linear map)**.** *Suppose $\lambda \in \mathbb{F}$ and $T \in \mathcal{L}(V,W)$. Then, $\mathcal{M}(\lambda T) = \lambda M(T)$.*

Proof. The matrix $\mathcal{M}(T)$ has entries $A_{j,k}$ that satisfy:

$$Tv_k = \sum_{j=1}^{m} A_{j,k}w_j$$

for each $j \in [m]$ and $k \in [n]$. Then, $\mathcal{M}(\lambda T)$ is given by:

$$\begin{aligned}(\lambda T)v_k &= \lambda \cdot Tv_k \\ &= \lambda \sum_{j=1}^{m} A_{j,k}w_j \\ &= \sum_{j=1}^{m} (\lambda A_{j,k})w_j\end{aligned}$$

So, $\mathcal{M}(\lambda T)$ has entries $\lambda A_{j,k}$ for each $j \in [m]$ and $k \in [n]$. Since $\lambda A_{j,k}$ are the entries of $\lambda\mathcal{M}(T)$, it follows $\mathcal{M}(\lambda T) = \lambda\mathcal{M}(T)$. □

Remark 3.2. For all positive integers m, n, the set of all m-by-n matrices with entries in $\mathbb{F}$ is denoted by $\mathbb{F}^{m,n}$. With the scalar multiplication and addition given in Definition 3.9, $\mathbb{F}^{m,n}$ is a vector space.

Theorem 3.14 (dimension of $\mathbb{F}^{m,n}$ is mn)**.** *Suppose that m, n are positive integers. With the addition and scalar multiplication in Definition 3.9, $\mathbb{F}^{m,n}$ is a vector space of dimension mn.*

Proof. It is clear that $\mathbb{F}^{m,n}$ is a vector space. Let E^l be the matrix defined so that $E^l_{j,k} = 1$ when $j = \lceil \frac{l-1}{n} \rceil$ and $k = ((l-1) \mod n) + 1$ and $E^l_{j,k} = 0$ otherwise for $l \in [mn]$. Then, each of the E^l are linearly independent, and for any matrix A, we can write $A = \sum_{j=1}^{m}\sum_{k=1}^{n} A_{j,k}E^{n(j-1)+k}$, so the E^l span $\mathbb{F}^{m,n}$. Hence they are a basis, so $\dim \mathbb{F}^{m,n} = mn$. □

Definition 3.10. Suppose A is a m-by-n matrix and B is a n-by-p matrix. Then, AB is defined to be the m-by-p matrix whose entry in row j, column k is given by the equation:

$$(AB)_{j,k} = \sum_{r=1}^{n} A_{j,r}B_{r,k}$$

The preceding definition makes sense because it allows $\mathcal{M}(ST) = \mathcal{M}(S)\mathcal{M}(T)$:

Theorem 3.15 (matrix of product of linear maps)**.** *If $T \in \mathcal{L}(U,V)$ and $S \in \mathcal{L}(V,W)$, then $\mathcal{M}(ST) = \mathcal{M}(S)\mathcal{M}(T)$.*

Proof. Let U have basis $u_1, ..., u_p$, V have basis $v_1, ..., v_n$, and W have basis $w_1, ..., w_m$. Let $\mathcal{M}(S) = A$ and $\mathcal{M}(T) = B$. Then for each $r \in [p]$:

$$\begin{aligned} ST u_r &= S(T u_r) \\ &= S\Big(\sum_{j=1}^{n} B_{j,r} v_j\Big) \\ &= \sum_{j=1}^{n} B_{j,r} S v_j \\ &= \sum_{j=1}^{n} B_{j,r} \Big(\sum_{k=1}^{m} A_{k,j} w_k\Big) \\ &= \sum_{k=1}^{m} \Big(\sum_{j=1}^{n} A_{k,j} B_{j,r}\Big) w_k \end{aligned}$$

So, row j, column k of $\mathcal{M}(ST)$ has the entry $\sum_{r=1}^{n} A_{j,r} B_{r,k}$. By Definition 3.10, this is the j,k entry of $\mathcal{M}(S)\mathcal{M}(T)$. Therefore, $\mathcal{M}(ST) = \mathcal{M}(S)\mathcal{M}(T)$. □

Definition 3.11 ($A_{j,\cdot}, A_{\cdot,k}$)**.** Let A be an m-by-n matrix.
If $1 \le j \le m$, then $A_{j,\cdot}$ denotes the 1-by-n matrix consisting of row j of A.
If $1 \le k \le n$, then $A_{\cdot,k}$ denotes the m-by-1 matrix consisting of column k of A.

When we take the product of a 1-by-n matrix and an n-by-1 matrix, the result is a 1-by-1 matrix. However, we usually denote this "singeleton" matrix as the number itself. This leads to the following:

Theorem 3.16 (entry of matrix product equals row times column)**.** *Suppose A is an m-by-n matrix and B is an n-by-p matrix. Then:*

$$(AB)_{j,k} = A_{j,\cdot} B_{\cdot,k}$$

Proof. This follows immediately from Definition 3.11 and Definition 3.10. □

Theorem 3.17 (column of matrix product equals matrix times column)**.** *Suppose A is an m-by-n matrix and B is an n-by-p matrix. Then*

$$(AB)_{\cdot,k} = A B_{\cdot,k}$$

Proof. We have the following equation based on Theorem 3.16:

$$(AB)_{1,k} = \sum_{r=1}^{n} A_{1,r}B_{r,k}$$

$$\vdots$$

$$(AB)_{m,k} = \sum_{r=1}^{n} A_{m,r}B_{r,k}$$

Since $r \in [n]$, the $B_{r,k}$ form the matrix $B_{\cdot,k}$. The $A_{j,r}$, $j \in [m]$ and $r \in [n]$, are the entries of A, so the above equations show that $(AB)_{\cdot,k} = AB_{\cdot,k}$. □

Theorem 3.18 (linear combination of columns)**.** *Suppose A is an m-by-n matrix and $b = \begin{bmatrix} b_1 \\ \vdots \\ b_n \end{bmatrix}$ is an n-by-1 matrix. Then:*

$$Ab = \sum_{k=1}^{n} b_k A_{\cdot,k}$$

Proof. Note that since b is an n-by-1 matrix, we write for row j, column k $b_{j,k} = b_{j,1} = b_j$ since there is only one column. Applying Theorem 3.17,

$$(Ab)_{\cdot,k} = Ab_{\cdot,k}$$

Note that Ab is an m-by-1 matrix, so $k = 1$ and for $1 \leq j \leq m$:

$$\begin{aligned}(Ab)_{j,1} &= \sum_{k=1}^{n} A_{j,k}b_{k,1} \\ &= \sum_{k=1}^{n} A_{j,k}b_k \\ &= \sum_{k=1}^{n} b_k A_{j,k}\end{aligned}$$

So, $(Ab)_{\cdot,1} = \sum_{k=1}^{n} b_k A_{\cdot,k}$ since the b_k are scalars. However, $(Ab)_{\cdot,1} = Ab$, so $Ab = \sum_{k=1}^{n} b_k A_{\cdot,k}$. □

Theorem 3.19 (matrix multiplication as linear combination of columns)**.** *Suppose C is an m-by-n matrix and R is a c-by-n matrix.*

(a) *If $k \in [n]$, then column k of CR is a linear combination of the columns of C with the coefficients of this linear combination coming from column k of R.*

(b) *If $j \in [m]$, then the row j of CR is a linear combination of the rows of R with the coefficients of this linear combination coming from row j of C.*

Proof. We formulate these in symbols first. We are asked to prove:

(a) $(CR)_{\cdot,k} = \sum_{j=1}^{c} R_{j,k} C_{\cdot,j}$.

(b) $(CR)_{j,\cdot} = \sum_{r=1}^{c} C_{j,r} R_{r,\cdot}$.

For item (a):

$$\begin{aligned} (CR)_{\cdot,k} &= CR_{\cdot,k} && \textit{Theorem } 3.17 \\ &= \sum_{j=1}^{c} (R_{\cdot,k})_j C_{\cdot,j} && \textit{Theorem } 3.18 \\ &= \sum_{j=1}^{c} R_{j,k} C_{\cdot,j} \end{aligned}$$

For item (b), we apply Definition 3.10 directly:

$$(CR)_{j,k} = \sum_{r=1}^{c} C_{j,r} R_{r,k}$$

The $R_{r,k}$ are the entries in $R_{r,\cdot}$, so $(CR)_{j,\cdot} = \sum_{r=1}^{c} C_{j,r} R_{r,\cdot}$.

□

Definition 3.12 (column rank, row rank)**.** Suppose A is an m-by-n matrix with entries in $\mathbb{F}$. The *column rank* is the dimension of the span of the columns of A in $\mathbb{F}^{m,1}$. The *row rank* of A is the dimension of the span of the rows of A in $\mathbb{F}^{1,n}$. Both the column and row rank of A are at most $\min(m, n)$.

Definition 3.13 (transpose)**.** The *transpose* of a $m - by - n$ matrix A, denoted by A^T, is the n-by-m matrix whose entries are defined as:

$$(A^T)_{k,j} = A_{j,k}$$

Theorem 3.20 (column-row factorization)**.** *Suppose A is an m-by-n matrix with entries in $\mathbb{F}$ and column rank $c \geq 1$. Then, there exist an m-by-c matrix C and a c-by-n matrix R (both with entries in $\mathbb{F}$) so that $A = CR$.*

Proof. The columns $A_{\cdot,1}, ..., A_{\cdot,n}$ are each in $\mathbb{F}^{m,1}$. We can reduce the list $A_{\cdot,1}, ..., A_{\cdot,n}$ into a basis of the span of the columns $A_{\cdot,1}, ..., A_{\cdot,n}$. The resulting basis has length c. The c columns in this basis can be concatenated into a larger

m-by-c matrix, whose columns span the same space as $A_{\cdot,1}, ..., A_{\cdot,n}$. Then, each $A_{\cdot,k}$ ($k \in [n]$) is a linear combination of the columns of C. Take the coefficients of this linear combination and put them into the kth column of a matrix R. Then, $A = CR$ by Theorem 3.19. □

Theorem 3.21 (column rank equals row rank)**.** *Suppose $A \in \mathbb{F}^{m,n}$. Then the column rank of A equals the row rank of A.*

Proof. By Theorem 3.20, $A = CR$ where C is an m-by-c matrix and R is a c-by-n matrix and c is the column rank. By Theorem 3.19, each row of A is a linear combination of the rows of R. Since there are c rows in R, the row rank of A is less than or equal to the column rank of A. To prove the other direction, we note that the column rank of A is equal to the row rank of A^T. Then, applying what we showed to A^T, the row rank of A^T is less than or equal to the column rank of A^T. However, the column rank of A^T is equal to the row rank of A. Therefore, the column rank of A is less than or equal to the row rank of A. That is, the column rank of A equals the row rank of A. □

The fact the column rank and row rank are equal implies the following definition:

Definition 3.14 (rank)**.** The rank of a matrix $A \in \mathbb{F}^{m,n}$ is the column rank of A.

3.4 Invertibility and Isomorphism

Definition 3.15 (invertible)**.** A linear map $T \in \mathcal{L}(V, W)$ is called *invertible* if there exists a linear map $S \in \mathcal{L}(W, V)$ such that ST equals the identity operator on V and TS equals the identity operator on W. S is called an *inverse* of T.

Theorem 3.22 (inverse is unique)**.** *An invertible linear map has a unique inverse.*

Proof. Let S_1 and S_2 be inverses of a linear map $T \in \mathcal{L}(V, W)$. We let I_V denote the identity operator in V and I_W denote the identity operator in W. Then, $S_1T = I_v$ and $S_2T = I_v$. Then, $S_1T = S_2T$. This implies $S_1TS_1 = S_2TS_1$. By associativity of linear operators, $S_1(TS_1) = S_2(TS_1)$. We know $TS_1 = I_w$, so $S_1I_w = S_2I_w$. Since $S_1I_w = S_1$ and $S_2I_w = S_2$, $S_1 = S_2$. □

Definition 3.16 (notation for inverse)**.** If T is invertible, we denote its inverse T^{-1}.

Theorem 3.23 (invertibility $\iff$ injectivity and surjectivity)**.** *A linear map is invertible if and only if it is injective and surjective.*

Proof. Let $T : V \to W$ be an invertible linear map. Suppose $Tu = Tv$ for some vectors $u, v \in V$. Since T is invertible, there exists T^{-1} so that $T^{-1}T = I$. Then, $T^{-1}(Tu) = T^{-1}(Tv)$, so $(T^{-1}T)u = (T^{-1}T)v$. Then, $u = v$. Therefore, T is injective. Let $w \in W$. Since $T^{-1} \in \mathcal{L}(W, V)$, we can put $v = T^{-1}w$, so

$Tv = T(T^{-1}w) = w$. Therefore, T is surjective.
Now suppose that T is injective and surjective. We define a map $S : W \to V$ so that $Sw = v$ whenever $w = Tv$. Since T is surjective, there exists $v \in V$ so that $w = Tv$ for each $w \in W$, so S is defined over all of W. It is well-defined because if $w = Tv_1$ and $w = Tv_2$ for $v_1, v_2 \in V$, then $v_1 = v_2$ because T is injective. It is also clear that S is a linear map. We now show it is the inverse. Let $v \in V$. Then, $STv = S(Tv) = v$ since $Tv \in W$. Therefore, $ST = I$. Let $w \in W$. Then, $TSw = T(Sw) = Tv$ where $w = Tv$. However, this means $TSw = w$. So, $TS = I$. Therefore, $S = T^{-1}$. □

Theorem 3.24 (injectivity is equivalent to surjectivity if $\dim V = \dim W < \infty$)**.** *Suppose that V and W are finite-dimensional vector spaces with $\dim V = \dim W$ and $T \in \mathcal{L}(V, W)$. Then the following conditions are equivalent:*

(i) T is invertible

(ii) T is injective

(iii) T is surjective

Proof. By Theorem 3.23, (i) implies (ii) and (i) implies (iii). We only need to show (ii) implies (iii) and (iii) implies (i). First, we show (ii) implies (iii). Let T be injective. Then, by the Theorem 3.7, $\dim V = \dim \operatorname{null} T + \dim \operatorname{range} T$. We know $\operatorname{null} T = \{0\}$ by Theorem 3.5. So, $\dim W = \dim V = \dim \operatorname{range} T$. Therefore, T is surjective. Now, we prove that (iii) implies (i). By Theorem 3.7, $\dim V = \dim \operatorname{null} T + \dim \operatorname{range} T$. Since $\dim \operatorname{range} T = \dim W = \dim V$, $\dim \operatorname{null} T = 0$. Then, $\operatorname{null} T = \{0\}$, so Theorem 3.5 implies that T is injective. Since T is injective and surjective, it is invertible by Theorem 3.23. □

Theorem 3.25 ($ST = I \iff TS = I$ on vector spaces of the same dimension)**.** *Suppose V and W are finite-dimensional vector spaces of the same dimension. Let $S \in \mathcal{L}(W, V)$ and $T \in \mathcal{L}(V, W)$. Then, $ST = I_V$ if and only if $TS = I_W$ where I_V and I_W are the identity maps on V and W, respectively.*

Proof. Suppose $ST = I_v$. If $v \in V$ and $Tv = 0$, then $v = I_V v = (ST)v = S(Tv) = S(0) = 0$. Therefore, $\operatorname{null} T = \{0\}$, so T is injective by Theorem 3.5. By Theorem 3.24, it is invertible since $\dim V = \dim W$. Therefore, if $ST = I_v$, then there exists T^{-1} so that $STT^{-1} = I_v T^{-1}$, so $S = T^{-1}$. This implies that $TS = I_W$. If we swap the roles of S and T, we get the other direction. □

Definition 3.17 (isomorphism)**.** An *isomorphism* is an invertible linear map. Two vector spaces are called an *isomorphism* if there is an isomorphism from one vector space into another.

Theorem 3.26 (dimension shows whether vector spaces are isomorphic)**.** *Two finite-dimensional vector spaces over $\mathbb{F}$ are isomorphic if and only if they have the same dimension.*

Proof. Let V and W be two finite-dimensional vector spaces over $\mathbb{F}$ with same dimension. Then, $v_1, ..., v_n$ is a basis for V and $w_1, ..., w_n$ is a basis for W. By the Lemma 3.1, we can define $T : V \to W$ so that $Tv_k = w_k$ for all $k \in [n]$. Suppose for some $v \in V$, $Tv = 0$. Then, $v = \sum_{k=1}^{n} c_k v_k$ for some $c_1, ..., c_n \in \mathbb{F}$. Therefore, $Tv = \sum_{k=1}^{n} c_k T v_k = \sum_{k=1}^{n} c_k w_k = 0$. Since $w_1, ..., w_n$ is linearly independent, $c_1 = ... = c_n = 0$. So, $v = 0$. Then, $\operatorname{null} T = \{0\}$. By Theorem 3.5, T is injective. By Theorem 3.24, T is invertible. Therefore, T is an isomorphism from V to W, so V and W are isomorphic. Now, suppose that V and W are isomorphic. There exists an invertible linear map $T : V \to W$. Since T is invertible, it is surjective, so $\operatorname{range} T = W$, and it is injective, so $\operatorname{null} T = \{0\}$ by Theorem 3.5. By Theorem 3.7, $\dim V = \dim \operatorname{null} T + \dim \operatorname{range} T = 0 + \dim W = \dim W$. □

Theorem 3.27 ($\mathcal{L}(V, W)$ and $\mathbb{F}^{m,n}$ are isomorphic)**.** *Suppose $v_1, ..., v_n$ is a basis of V and $w_1, ..., w_m$ is a basis of W. Then, $\mathcal{M}$ is an isomorphism between $\mathcal{L}(V, W)$ and $\mathbb{F}^{m,n}$.*

Proof. Definition 3.9 implies that $\mathcal{M}$ is a linear map. To show it is invertible, let $A \in \mathbb{F}^{n,m}$. By Lemma 3.1, we can define $T : V \to V$ so that $Tv_k = \sum_{j=1}^{m} A_{j,k} w_j$ for each $k \in [n]$ since $v_1, ..., v_n$ is a basis. Then, A is the matrix of the linear transformation T, so $A = \mathcal{M}(T)$. Therefore, $\mathcal{M}$ is surjective. Now, we show $\mathcal{M}$ is injective. Suppose that $\mathcal{M}(T) = 0$ (the zero matrix). Then, this implies $Tv_k = \sum_{j=1}^{m} 0 \cdot w_j = 0$ for all $k \in [n]$. Since $v_1, ..., v_n$ is a basis, it cannot be that $v_k = 0$ for any of the $k \in [n]$. So, $T = 0$. Since $\operatorname{null} \mathcal{M} = \{0\}$, $\mathcal{M}$ is injective by Theorem 3.5. By Theorem 3.23, $\mathcal{M}$ is invertible. Therefore, $\mathcal{M}$ is an isomorphism. □

Theorem 3.28 ($\dim \mathcal{L}(V, W) = (\dim V)(\dim W)$)**.** *Suppose V and W are finite-dimensional. Then, $\mathcal{L}(V, W)$ is finite-dimensional and $\dim \mathcal{L}(V, W) = (\dim V)(\dim W)$.*

Proof. Let $\dim V = n$ and $\dim W = m$. Then, $\mathcal{L}(V, W)$ and $\mathbb{F}^{n,m}$ are isomorphic by Theorem 3.27. By Theorem 3.26, $\dim \mathcal{L}(V, W) = \dim \mathbb{F}^{n,m} = nm = (\dim V)(\dim W)$. □

Definition 3.18 (matrix of a vector, $\mathcal{M}(v)$)**.** Suppose $v \in V$ and $v_1, ..., v_n$ is a basis of V. The *matrix of v* with respect to this basis is the n-by-1 matrix:

$$\mathcal{M}(v) = \begin{bmatrix} b_1 \\ \vdots \\ b_n \end{bmatrix}$$

where $b_1, ..., b_n \in \mathbb{F}$ satisfy $v = \sum_{k=1}^{n} b_k v_k$.

Theorem 3.29 ($\mathcal{M}(T)_{\cdot,k} = \mathcal{M}(Tv_k)$)**.** *Suppose $T \in \mathcal{L}(V, W)$ and $v_1, ..., v_n$ is a basis of V and $w_1, ..., w_m$ is a basis of W. Let $1 \leq k \leq n$. Then, the kth column of $\mathcal{M}(T)$, which is denoted by $\mathcal{M}(T)_{\cdot,k}$, equals $\mathcal{M}(Tv_k)$.*

Proof. Recall that $A = \mathcal{M}(T)$ means $Tv_k = \sum_{j=1}^{m} A_{j,k} w_j$ for each $k \in [n]$. By Definition 3.18, $\mathcal{M}(Tv_k) = \begin{bmatrix} A_{1,k} \\ \vdots \\ A_{m,k} \end{bmatrix}$. This is exactly equal to $\mathcal{M}(T)_{\cdot,k}$. □

Theorem 3.30 (linear maps act like matrix multiplication)**.** *Suppose $T \in \mathcal{L}(V, W)$ and $v \in V$. Suppose $v_1, ..., v_n$ is a basis of V and $w_1, ..., w_m$ is a basis for W. Then, $\mathcal{M}(Tv) = \mathcal{M}(T)\mathcal{M}(v)$.*

Proof. First, if we let $A = \mathcal{M}(T)$:

$$\begin{aligned} Tv &= T\Big(\sum_{k=1}^{n} c_k v_k\Big) \\ &= \sum_{k=1}^{n} c_k \cdot Tv_k \\ &= \sum_{k=1}^{n} c_k \sum_{j=1}^{m} A_{j,k} w_j \\ &= \sum_{j=1}^{m} \Big(\sum_{k=1}^{n} c_k A_{j,k}\Big) w_j \end{aligned}$$

Since $Tv \in \mathbb{F}^{m,1}$, it follows that:

$$\mathcal{M}(Tv) = \begin{bmatrix} \sum_{k=1}^{n} c_k A_{1,k} \\ \vdots \\ \sum_{k=1}^{n} c_k A_{m,k} \end{bmatrix}$$

Since $\mathcal{M}(T) = A$ and $\mathcal{M}(v) = \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix}$, Definition 3.10 implies that the entry in row j, column k of $\mathcal{M}(T)\mathcal{M}(v)$ is given by:

$$(\mathcal{M}(T)\mathcal{M}(v))_{j,k} = \sum_{r=1}^{n} A_{j,r} c_{r,k}$$

Since $\mathcal{M}(T)\mathcal{M}(v) \in \mathbb{F}^{m,1}$, there is only one column, so we can instead write:

$$\begin{aligned} (\mathcal{M}(T)\mathcal{M}(v))_j &= \sum_{r=1}^{n} A_{j,r} c_r \\ &= \sum_{k=1}^{n} c_k A_{j,k} \end{aligned}$$

where the last line is just a change-of-variable. Since $(\mathcal{M}(Tv))_j = \sum_{k=1}^{n} c_k A_{j,k}$, $\mathcal{M}(Tv) = \mathcal{M}(T)\mathcal{M}(v)$. □

Theorem 3.31 (dimension of range T equals column rank of $\mathcal{M}(T)$)**.** *Suppose V and W are finite-dimensional and $T \in \mathcal{L}(V,W)$. Then,* $\dim \operatorname{range} T$ *is equal to the column rank of $\mathcal{M}(T)$.*

Proof. Suppose $v_1, ..., v_n$ is a basis for V and $w_1, ..., w_m$ is a basis for W. The map that takes $w \in W$ to $\mathcal{M}(w) \in \mathbb{F}^{m,1}$ is an isomorphism. If we restrict this map to $\operatorname{range} T = \operatorname{span}(Tv_1, ..., Tv_n)$ (see Exercise 10 in section 3B of book), the map is an isomorphism from $\operatorname{range} T$ to $\operatorname{span}(\mathcal{M}(Tv_1), ..., \mathcal{M}(Tv_n))$. From Theorem 3.29, we know that $\mathcal{M}(Tv_k) = \mathcal{M}(T)_{\cdot,k}$ for each $k \in [n]$, so $\dim \operatorname{range} T =$ column rank $\mathcal{M}(T)$ by Theorem 3.26. □

Definition 3.19 (identity matrix)**.** Suppose n is a positive integer. The n-by-n matrix:

$$\begin{bmatrix} 1 & ... & 0 \\ \vdots & \ddots & \vdots \\ 0 & ... & 1 \end{bmatrix}$$

with 1's on the diagonal and 0's elsewhere is called the *identity matrix* and is denoted by I.

Definition 3.20 (invertible matrix)**.** A square matrix A is called *invertible* if there is a square matrix B of the same size such that $AB = BA = I$. We call B the *inverse* of A and denote it by A^{-1}. The words *singular* and *nonsingular* are also used to call A invertible or noninvertible, respectively.

Theorem 3.32 (matrix of product of linear maps with basis emphasized)**.** *Suppose $T \in \mathcal{L}(U,V)$ and $S \in \mathcal{L}(V,W)$. If $u_1, ..., u_m$ is a basis of U and $v_1, ..., v_n$ is a basis of V, and $w_1, ..., w_p$ is a basis of W, then:*

$$\mathcal{M}(ST, (u_1, ..., u_m), (w_1, ..., w_p)) = \mathcal{M}(S, (v_1, ..., v_n), (w_1, ..., w_p))\mathcal{M}(T, (u_1, ..., u_m), (v_1, ..., v_n))$$

Proof. See Theorem 3.15. □

Theorem 3.33 (matrix of identity operator with respect to two bases)**.** *Suppose that $u_1, ..., u_n$ and $v_1, ..., v_n$ are bases of V. Then, the matrices $\mathcal{M}(I, (u_1, ..., u_n), (v_1, ..., v_n))$ and $\mathcal{M}(I, (v_1, ..., v_n), (u_1, ..., u_n))$ are invertible and are inverses of each other.*

Proof. We can write:

$$I = \mathcal{M}(I, (v_1, ..., v_n), (u_1, ..., u_n))\mathcal{M}(I, (u_1, ..., u_n), (v_1, ..., v_n))$$

And also:

$$I = \mathcal{M}(I, (u_1, ..., u_n), (v_1, ..., v_n))\mathcal{M}(I, (v_1, ..., v_n), (u_1, ..., u_n))$$

Equating the two proves the result. □

Theorem 3.34 (change-of-basis formula)**.** *Suppose $T \in \mathcal{L}(V)$. Suppose $u_1, ..., u_n$ and $v_1, ..., v_n$ are two bases for V. Let $A = \mathcal{M}(T, (u_1, ..., u_n))$ and $B = \mathcal{M}(T, (v_1, ..., v_n))$. Let $C = \mathcal{M}(I, (u_1, ..., u_n), (v_1, ..., v_n))$. Then, $A = C^{-1}BC$.*

Proof. We first write down the matrices involved:

$$\begin{aligned} A &= \mathcal{M}(T, (u_1, ..., u_n)) \\ B &= \mathcal{M}(T, (v_1, ..., v_n)) \\ C &= \mathcal{M}(I, (u_1, ..., u_n), (v_1, ..., v_n)) \\ C^{-1} &= \mathcal{M}(I, (v_1, ..., v_n), (u_1, ..., u_n)) \qquad \text{Theorem 3.33} \end{aligned}$$

Then:

$$\begin{aligned} A &= \mathcal{M}(IT, (u_1, ..., u_n)) \\ &= \mathcal{M}(I, (v_1, ..., v_n), (u_1, ..., u_n))\mathcal{M}(T, (u_1, ..., u_n), (v_1, ..., v_n)) \qquad \text{Theorem 3.32} \\ &= C^{-1}\mathcal{M}(T, (u_1, ..., u_n), (v_1, ..., v_n)) \\ &= C^{-1}\mathcal{M}(T, (v_1, ..., v_n), (v_1, ..., v_n))\mathcal{M}(I, (u_1, ..., u_n), (v_1, ..., v_n)) \quad \text{Theorem 3.32} \\ &= C^{-1}BC \end{aligned}$$

□

Theorem 3.35 (matrix of inverse equals inverse of matrix)**.** *Suppose that $v_1, ..., v_n$ is a basis of V and $T \in \mathcal{L}(V)$ is invertible. Then $\mathcal{M}(T^{-1}) = (\mathcal{M}(T))^{-1}$, where both matrices are with respect to the basis $v_1, ..., v_n$.*

Proof. We know that $TT^{-1} = T^{-1}T = I$ by Definition 3.15. Then, $\mathcal{M}(TT^{-1}) = \mathcal{M}(I) = I$, the identity matrix. By Theorem 3.32, $I = \mathcal{M}(T)\mathcal{M}(T^{-1})$. The same logic applies to $T^{-1}T = I$, so $I = \mathcal{M}(T^{-1})\mathcal{M}(T)$. Then, $I = \mathcal{M}(T)\mathcal{M}(T^{-1}) = \mathcal{M}(T^{-1})\mathcal{M}(T)$, so by Definition 3.20, $\mathcal{M}(T^{-1})$ is the inverse of $\mathcal{M}(T)$, i.e, $(\mathcal{M}(T))^{-1} = \mathcal{M}(T^{-1})$. □

3.5 Products and Quotients of Vector Spaces

Definition 3.21 (product of vector spaces)**.** Suppose $V_1, ..., V_m$ are vector spaces over $\mathbb{F}$. The *product* $V_1 \times ... \times V_m = \{(v_1, ..., v_m) : v_1 \in V_1, ..., v_m \in V_m\}$. We define addition over $V_1 \times ... \times V_m$ by:

$$(u_1, ..., u_m) + (v_1, ..., v_m) = (u_1 + v_1, ..., u_m + v_m)$$

And we define scalar multiplication by:

$$\lambda(v_1, ..., v_m) = (\lambda v_1, ..., \lambda v_m)$$

Theorem 3.36 (product of vector spaces is a vector spaces)**.** *Suppose $V_1, ..., V_m$ are vector spaces over $\mathbb{F}$. Then, $V_1 \times ... \times V_m$ is a vector space over $\mathbb{F}$.*

Proof. We prove all the properties in Definition 1.2:

1. Commutativity: $(u_1, ..., u_m) + (v_1, ..., v_m) = (u_1 + v_1, ..., u_m + v_m) = (v_1 + u_1, ..., v_m + u_m)$.

2. Associativity:

$$\begin{aligned}((u_1,...,u_m)+(v_1,...,v_m))+(w_1,...,w_m) &= (u_1+v_1,...,u_m+v_m)+(w_1,...,w_m)\\ &= ((u_1+v_1)+w_1,...,(u_m+v_m)+w_m)\\ &= (u_1+(v_1+w_1),...,u_m+(v_m+w_m))\\ &= (u_1,...,u_m)+(v_1+w_1,...,v_m+w_m)\\ &= (u_1,...,u_m)+((v_1,...,v_m)+(w_1,...,w_m))\end{aligned}$$

3. Additive identity: $(u_1,...,u_m)+(0,...,0)=(u_1,...,u_m)$ where $(0,...,0)$ is the element where we choose the zero vector from each of the m vector spaces.

4. Additive inverse: $(u_1,...,u_m)+(-u_1,...,-u_m)=(0,...,0)$.

5. Multiplicative identity: By how scalar multiplication is defined, $1(u_1,...,u_m)=(1\cdot u_1,...,1\cdot u_m)=(u_1,...,u_m)$.

6. Distributive property: Let $a\in\mathbb{F}$. Then, $a((u_1,...,u_m)+(v_1,...,v_m))=a(u_1+v_1,...,u_m+v_m)=(a(u_1+v_1),...,a(u_m+v_m))=(au_1+av_1,...,au_m+av_m)=(au_1,...,au_m)+(av_1,...,av_m)=a(u_1,...,u_m)+a(v_1,...,v_m)$. Let $b\in\mathbb{F}$. Then, $(a+b)(u_1,...,u_m)=((a+b)u_1,...,(a+b)u_m)=(au_1+bu_1,...,au_m+bu_m)=(au_1,...,au_m)+(bu_1,...,bu_m)=a(u_1,...,u_m)+b(u_1,...,u_m)$.

□

Theorem 3.37 (dimension of a product is the sum of dimensions)**.** *Suppose $V_1,...,V_m$ are finite-dimensional vector spaces. Then, $V_1\times...\times V_m$ is finite-dimensional and:*

$$\dim\prod_{k=1}^{m}V_k=\sum_{k=1}^{m}\dim V_k$$

Proof. We will use v_{kj} to denote the jth basis vector for V_k. Since each V_k is finite-dimensional, each has a list of basis vector of size n_k. Consider any $(v_1,...,v_m)\in\prod_{k=1}^{m}V_k$. Then, we can write each $v_k=\sum_{j=1}^{n_k}c_{kj}v_{kj}$, So, we can write $(v_1,...,v_m)=\sum_{k=1}^{m}\sum_{j=1}^{n_k}c_{kj}e_{kj}$ where e_{kj} is the m -tuple of with zeros everywhere except at position kj, where it is equal to v_{kj}. Clearly, e_{kj} spans $\prod_{k=1}^{m}V_k$. They are also linearly independent. So they form a basis. Since there are n_k basis vectors for each $k\in[m]$, the basis has length $\sum_{k=1}^{m}n_k=\sum_{k=1}^{m}\dim V_k$, so $\dim\prod_{k=1}^{m}V_k=\sum_{k=1}^{m}\dim V_k$. □

Theorem 3.38 (products and direct sums)**.** *Suppose that $V_1,...,V_m$ are subspaces of V. Define a linear map $\Gamma:V_1\times...\times V_m\to V_1+...+V_m$ by:*

$$\Gamma(v_1,...,v_m)=\sum_{k=1}^{m}v_k$$

Then, $V_1 + ... + V_m$ is a direct sum iff Γ is injective.

Proof. Suppose $V_1+...+V_m$ is a direct sum. Suppose $\Gamma(v_1, ..., v_m) = \Gamma(u_1, ..., u_m)$. Then, $\sum_{k=1}^m v_k = \sum_{k=1}^m u_k$. By Definition 1.5, there is only one way to write $v = \sum_{k=1}^m v_k$, so this means $v_k = u_k$ for every $k \in [m]$. Therefore, Γ is injective. Now, suppose that Γ is injective. Suppose $\sum_{k=1}^m v_k = 0$. Then, $\Gamma(v_1, ..., v_m) = 0$ only when $v_1 = ... = v_m = 0$ by Theorem 3.5. By Theorem 1.8, $V_1 + ... + V_m$ is a direct sum. □

Remark 3.3. We know that Γ is surjective since every element in $v \in \sum_{k=1}^m V_k$ can be expressed as $v = \sum_{k=1}^m v_k$ where $v_k \in V_k$ for each $k \in [m]$; $\Gamma(v_1, ..., v_m) = v$. However, given the result from the preceding theorem, if Γ is also injective, then Γ is invertible by Theorem 3.23.

Theorem 3.39 (a sum is a direct sum iff dimensions add up)**.** *Suppose V is finite-dimensional and $V_1, ..., V_m$ are subspaces of V. Then $V_1 + ... + V_m$ is a direct sum iff* $\dim \sum_{k=1}^m V_k = \sum_{k=1}^m \dim V_k$.

Proof. Suppose that $\dim \sum_{k=1}^m V_k = \sum_{k=1}^m \dim V_k$. Define $\Gamma : \prod_{k=1}^m V_k \to \sum_{k=1}^m V_k$ so that $\Gamma(v_1, ..., v_m) = v_1+...+v_m$. We know this map is surjective. By Theorem 3.37, $\dim \prod_{k=1}^m V_k = \sum_{k=1}^m \dim V_k$, so $\dim \prod_{k=1}^m V_k = \dim \sum_{k=1}^m V_k$. So Theorem 3.24, Γ is injective. By Theorem 3.38, since Γ is injective, $\sum_{k=1}^m V_k$ is a direct sum.
Now suppose that $\sum_{k=1}^m V_k$ is a direct sum. Then, by Theorem 3.38, Γ is injective. Since it is also surjective, it is invertible. Then, Γ is an isomorphism. Hence, $\dim \prod_{k=1}^m V_k = \dim \sum_{k=1}^m V_k$ by Theorem 3.26. By Theorem 3.37, $\dim \prod_{k=1}^m V_k = \sum_{k=1}^M \dim V_k$, so $\dim \sum_{k=1}^m V_k = \sum_{k=1}^m \dim V_k$. □

Now we move on to quotient spaces.

Definition 3.22 (translates)**.** Suppose $v \in V$ and $U \subseteq V$. Then, $v + U := \{v + u : u \in U\}$. The set $v + U$ is called a *translate* of U.

Definition 3.23 (quotient space)**.** Suppose U is a subspace of V. Then the *quotient space* V/U is the set of all translates of U, i.e, $V/U = \{v + U : v \in V\}$.

Theorem 3.40 (two translates of a subspace are equal or disjoint)**.** *Suppose U is a subspace of V and $v, w \in V$. Then the following are equivalent:*

(i) $v - w \in U$

(ii) $v + U = w + U$

(iii) $(v + U) \cap (w + U) \neq \emptyset$

Proof. It is clear that (ii) implies (iii). We prove (i) implies (ii) and (iii) implies (i). Suppose $v-w \in U$. For any $u \in U$, $v+u = v+u+(w-w) = w+(v-w+u)$. Since $u \in U$ and $v - w \in U$, $v - w + u \in U$, so $v + u \in w + U$. Similarly, $w + u = w + u + (v - v) = v + (w - v + u)$. Since $v - w \in U$, $w - v \in U$, so $w - v + u \in U$. Therefore, $w + u \in v + U$. Therefore, $v + U = w + U$. Now, we show (iii) implies (i). Suppose that $s \in (v + U) \cap (w + U)$. Then, $s = v + u_1$ and $s = w + u_2$ for some $u_1, u_2 \in U$. Therefore, $v - w = u_2 - u_1 \in U$. □

Definition 3.24 (addition and scalar multiplication on V/U)**.** Suppose U is a subspace of V. Then addition and scalar multiplication are defined on V/U as follows:

(i) $(v + U) + (w + U) = (v + w) + U$

(ii) $\lambda(v + U) = (\lambda v) + U$

for all $v, w \in V$ and $\lambda \in \mathbb{F}$

Theorem 3.41 (quotient space is a vector space)**.** *Suppose U is a subspace of V. Then, V/U, with addition and scalar multiplication defined as in Definition 3.24, is a vector space.*

Proof. First, we show that the definition Definition 3.24 makes sense. Let $v_1, v_2, w_1, w_2 \in V$. Suppose $v_1 + U = v_2 + U$ and $w_1 + U = w_2 + U$. We need to show $(v_1 + w_1) + U = (v_2 + w_2) + U$. By Theorem 3.40, $v_1 - v_2 \in U$ and $w_1 - w_2 \in U$. Since U is a subspace, it is closed under addition, so $(v_1 + w_1) - (v_2 + w_2) = (v_1 - v_2) + (w_1 - w_2) \in U$. By Theorem 3.40, that means $(v_1 + w_1) + U = (v_2 + w_2) + U$. Similarly, if $v_1 + U = v_2 + U$ and $\lambda \in \mathbb{F}$, then since $v_1 - v_2 \in U$, $\lambda(v_1 - v_2) = \lambda v_1 - \lambda v_2 \in U$, so $\lambda v_1 + U = \lambda v_2 + U$. Therefore, $\lambda v_1 + U = \lambda v_2 + U$, so $\lambda(v_1 + U) = \lambda(v_2 + U)$. We show the properties in Definition 1.2:

(i) Commutativity:

$$\begin{aligned}(v + U) + (w + U) &= (v + w) + U \\ &= (w + u) + U \\ &= (w + U) + (v + U)\end{aligned}$$

(ii) Associativity:

$$\begin{aligned}((v_1 + U) + (v_2 + U)) + (v_3 + U) &= ((v_1 + v_2) + U) + (v_3 + U) \\ &= ((v_1 + v_2) + v_3) + U \\ &= (v_1 + (v_2 + v_3)) + U \\ &= (v_1 + U) + ((v_2 + v_3) + U) \\ &= (v_1 + U) + ((v_2 + U) + (v_3 + U))\end{aligned}$$

(iii) Additive identity: $(v + U) + (0 + U) = (v + 0) + U = v + U$

(iv) Additive inverse: $(v + U) + ((-v) + U) = (v + (-v)) + U = 0 + U$

(v) Multiplicative identity: $1(v + U) = (1v) + U = v + U$

(vi) Distributive Property: if $a, b \in \mathbb{F}$, then $a((v+U)+(w+U)) = a((v+w)+U) = a(v+w)+U = (av+aw)+U = (av+U)+(aw+U) = a(v+U)+a(w+U)$. And $(a+b)(v+U) = ((a+b)v)+U = (av+bv)+U = (av+U)+(bv+U) = a(v+U)+b(v+U)$.

□

Definition 3.25 (quotient map)**.** Suppose U is a subspace of V. The *quotient map* $\pi : V \to V/U$ is the linear map defined by $\pi(v) = v + U$ for each $v \in V$.

Remark 3.4. We show the quotient map is linear. Suppose $v_1, v_2 \in V$. Then, $\pi(v_1 + v_2) = (v_1 + v_2) + U = (v_1 + U) + (v_2 + U) = \pi(v_1) + \pi(v_2)$. Let $\lambda \in \mathbb{F}$. Then, $\pi(\lambda v) = (\lambda v) + U = \lambda(v + U) = \lambda\pi(v)$.

Theorem 3.42 (dimension of quotient space)**.** *Suppose V is finite-dimensional and U is a subspace of V. Then:* $\dim V/U = \dim V - \dim U$

Proof. We will use Theorem 3.7 since the quotient map $\pi : V \to V/U$ is linear. It is clear that π is surjective, so $\operatorname{range}\pi = V/U$. We show that $\operatorname{null}\pi = U$. Suppose that $u \in U$. Then, $u - 0 \in U$. By Theorem 3.40, $u + U = 0 + U$. That is, $\pi(u) = 0 + U$ for all $u \in U$. So, $U \subset \operatorname{null}\pi$. Suppose now that $v \in \operatorname{null}\pi$. Then, $\pi(v) = 0 + U$. That is, $v + U = 0 + U$. By Theorem 3.40, $v = v - 0 \in U$, so $\operatorname{null}\pi = U$. Using Theorem 3.7, $\dim V = \dim\operatorname{null}\pi + \dim\operatorname{range}\pi = \dim U + \dim V/U$, meaning $\dim V/U = \dim V - \dim U$. □

Definition 3.26 ($\widetilde{T}$)**.** Suppose $T \in \mathcal{L}(V, W)$. Define $\widetilde{T} : V/(\operatorname{null}T) \to W$ by $\widetilde{T}(v + \operatorname{null}T) = Tv$.

Remark 3.5. We show that $\widetilde{T}$ is well-defined and linear. First, suppose $v_1 + \operatorname{null}T = v_2 + \operatorname{null}T$. We need to show that $Tv_1 = Tv_2$. By Theorem 3.40, $v_1 - v_2 \in \operatorname{null}T$. That is, $T(v_1 - v_2) = 0$. Therefore, $Tv_1 = Tv_2$ (where we used the linearity of T). So, $\widetilde{T}$ is well-defined. It is linear since T is linear.

Theorem 3.43 (null space and range of $\widetilde{T}$)**.** *Suppose $T \in \mathcal{L}(V, W)$. Then:*

(a) $\widetilde{T} \circ \pi = T$*, where π is the quotient-map of V onto $V/(\operatorname{null}T)$.*

(b) $\widetilde{T}$ *is injective.*

(c) $\operatorname{range}\widetilde{T} = \operatorname{range}T$.

(d) $V/(\operatorname{null}T)$ *and* $\operatorname{range}T$ *are isomorphic vector spaces.*

Proof.

(a) Let $v \in V$. Then, $(\widetilde{T} \circ \pi)(v) = \widetilde{T}(\pi(v)) = \widetilde{T}(v + \operatorname{null}T) = Tv$. Therefore, $\widetilde{T} \circ \pi = T$.

(b) Suppose $\widetilde{T}(v + \operatorname{null}T) = \widetilde{T}(u + \operatorname{null}T)$. Then, $Tv = Tu$, which implies $T(u - v) = 0$. Hence, $u - v \in \operatorname{null}T$. By Theorem 3.40, this means that $u + \operatorname{null}T = v + \operatorname{null}T$.

(c) We can write $\operatorname{range} T = \{Tv : v \in V\} = \{\widetilde{T}(v + \operatorname{null} T) : v \in V\}$. Since every element in $V/(\operatorname{null} T)$ can be written as $v + \operatorname{null} T$ for some $v \in V$, it follows that $\operatorname{range} T = \operatorname{range} \widetilde{T}$.

(d) Let $S : V/(\operatorname{null} T) \to \operatorname{range} T$ be the map defined so that $S(v + \operatorname{null} T) = \widetilde{T}(v + \operatorname{null} T)$. Since $\widetilde{T}$ is injective by part (b), we know that S is also injective. Also, S is surjective since $\operatorname{range} T = \operatorname{range} \widetilde{T}$, so for any $w \in \operatorname{range} T$, $w = Tv$, and $S(v + \operatorname{null} T) = Tv = w$. By Theorem 3.23, S is invertible. Since it is also a linear map, S is an isomorphism.

□

3.6 Duality

Definition 3.27 (linear functional)**.** A *linear functional* on V is a linear map from V onto $\mathbb{F}$. In other words, a linear functional is an element of $\mathcal{L}(V, \mathbb{F})$.

Definition 3.28 (dual space)**.** The *dual space* of V, denoted by V', is the vector space of all linear functionals on V. In other words, $V' = \mathcal{L}(V, F)$.

Theorem 3.44 ($\dim V' = \dim V$)**.** *Suppose V is finite-dimensional. Then, V' is also finite-dimensional and* $\dim V' = \dim V$.

Proof. Since $V' = \mathcal{L}(V, \mathbb{F})$ and $\mathbb{F}$ is a vector space of dimension 1, Theorem 3.28 implies $\dim V' = \dim \mathcal{L}(V, \mathbb{F}) = (\dim V)(\dim \mathbb{F}) = \dim V$. □

Definition 3.29 (dual basis)**.** If $v_1, ..., v_n$ is a basis of V, then the *dual basis* of $v_1, ..., v_n$ is the list $\varphi_1, ..., \varphi_n$ of elements of V', where each φ_j is the linear functional on V such that:

$$\varphi_j(v_k) = \begin{cases} 0 & k \neq j \\ 1 & k = j \end{cases}$$

Remark 3.6. We show that φ_j is well-defined for each $j \in [n]$ and is a basis. By Lemma 3.1, each of the basis vectors in V are mapped to an element of $\mathbb{F}$ (in this case, 0 or 1) by each φ_j, so φ_j is a linear map (hence well-defined). Next, we show it is a basis for V'. First, Theorem 3.44 tells us that V' has the same dimension as V, so the list $\varphi_1, ..., \varphi_n$ has the same length has the basis $v_1, ..., v_n$. Next, we let $T \in \mathcal{L}(V, \mathbb{F})$. Let $v \in V$. Then, $v = \sum_{k=1}^{n} c_k v_k$ for some coefficients $c_1, ..., c_n \in \mathbb{F}$. We have that:

$$\begin{aligned} \varphi_j v &= \varphi_j\Big(\sum_{k=1}^{n} c_k v_k\Big) \\ &= \sum_{k=1}^{n} c_k \varphi_j(v_k) \\ &= c_j \end{aligned}$$

for each $j \in [n]$. Then, we can write:

$$\begin{aligned} Tv &= T\Big(\sum_{k=1} c_k v_k\Big) \\ &= \sum_{k=1}^{n} c_k (Tv_k) \\ &= \sum_{k=1}^{n} (\varphi_k v)(Tv_k) \end{aligned}$$

Put $b_k = Tv_k \in \mathbb{F}$ for each $k \in [n]$. Then:

$$\begin{aligned} Tv &= \sum_{k=1}^{n} (\varphi_k v) b_k \\ &= \Big(\sum_{k=1}^{n} b_k \varphi_k\Big) v \end{aligned}$$

So, $T = \sum_{k=1}^{n} b_k \varphi_k$. So, $\mathcal{L}(V, \mathbb{F}) = \text{span}(\varphi_1, ..., \varphi_n)$. By Theorem 2.13, $\varphi_1, ..., \varphi_n$ is a basis for V'.

Definition 3.30 (dual map)**.** Suppose $T \in \mathcal{L}(V, W)$. The *dual map* of T is the linear map $T' \in \mathcal{L}(W', V')$ defined for each $\varphi \in W'$ by:

$$T'(\varphi) = \varphi \circ T$$

Theorem 3.45 (algebraic properties of dual maps)**.** *Suppose $T \in \mathcal{L}(V, W)$. Then:*

(a) $(S + T)' = S' + T'$ *for all* $S \in \mathcal{L}(V, W)$.

(b) $(\lambda T)' = \lambda T'$ *for all* $\lambda \in \mathbb{F}$.

(c) $(ST)' = T'S'$ *for all* $S \in \mathcal{L}(W, U)$.

Proof.

(a) Let $\varphi \in W'$. Then, $(S+T)'(\varphi) = \varphi(S+T) = \varphi S + \varphi T = S'(\varphi) + T'(\varphi) = (S' + T')(\varphi)$.

(b) Let $\varphi \in W'$. Then, $(\lambda T)'(\varphi) = \varphi(\lambda T) = \lambda(\varphi T) = \lambda T'(\varphi)$.

(c) Let $\varphi \in W'$. Then, $(T'S')(\varphi) = T'(S'(\varphi)) = T'(\varphi S) = (\varphi S)(T) = \varphi(ST) = (ST)'$.

□

Remark 3.7. In Theorem 3.45, items (a) and (b) imply that the function taking T to T' is a linear map from $\mathcal{L}(V,W)$ to $\mathcal{L}(W',V')$. Also, sometimes the dual space is denoted as V^* instead of V'.

Definition 3.31 (annihilator)**.** For $U \subseteq V$, the *annihilator* of U, denoted by U^0, is defined by:

$$U^0 = \{\varphi \in V' : \varphi(u) = 0 \text{ for all } u \in U\}$$

Example 3.1. Suppose U is a subspace of $\mathcal{P}(\mathbb{R})$, consisting of polynomial multiples of x^2. If φ is the linear functional on $\mathcal{P}(\mathbb{R})$ defined by $\varphi(p) = p'(0)$, then $\varphi \in U^0$ (since $(cx^2)' = 2xc$ which is zero when $x = 0$).

Theorem 3.46 (the annihilator is a subspace)**.** *Let $U \subset V$. Then U^0 is a subspace of V'.*

Proof. Suppose $\varphi_1, \varphi_2 \in U^0$. Then, for every $u \in U$, $\varphi_1(u) = 0$ and $\varphi_2(u) = 0$. Therefore, for every $u \in U$, $(\varphi_1 + \varphi_2)(u) = \varphi_1(u) + \varphi_2(u) = 0$. That is, $\varphi_1 + \varphi_2 \in U^0$. Now, let $\lambda \in \mathbb{F}$ and $\varphi \in U^0$. Then, $(\lambda\varphi)(u) = \lambda(\varphi(u)) = 0$ for every $u \in U$ since φ is a linear map. Therefore, $\lambda\varphi \in U$. It is clear that the zero map $0 : V \to \mathbb{F}$ is in U^0. So, U^0 is a subspace of V'. □

Theorem 3.47 (dimension of the annihilator)**.** *Suppose V is finite-dimensional and U is a subspace of V. Then,* $\dim U^0 = \dim V - \dim U$.

Proof. Let $i \in \mathcal{L}(U,V)$ be the inclusion map defined by $i(u) = u$ for all $u \in U$. Thus, i' is a linear map from V' to U'. By Theorem 3.44, $\dim V = \dim V'$, and by Theorem 3.7:

$$\dim V = \dim V' = \dim \operatorname{null} i' + \dim \operatorname{range} i'$$

Let $\varphi \in \operatorname{null} i'$. Then, $i'(\varphi) = \varphi \circ i = 0$. That is, for every $u \in U$, $\varphi(i(u)) = 0$. Since $i(u) = u$ for all $u \in U$, $\varphi u = 0$ for all $u \in U$, so $|varphi \in U^0$. Therefore, $\operatorname{null} i' \subset U^0$. Now, suppose $\varphi \in U^0$. Then, $\varphi u = 0$ for all $u \in U$. Since $i(u) = u$ for all $u \in U$, $(\varphi \circ i)(u) = 0$ for all $u \in U$. That is, $i'(\varphi) = \varphi \circ i = 0$. So, $\varphi \in \operatorname{null} i'$, meaning $U^0 \subset \operatorname{null} i'$. Therefore, $\operatorname{null} i' = U^0$. Now, let $\varphi \in U'$. Then, by Exercise 13 in section 3A, φ can be extended to a linear functional ψ on V. Then, $i'(\psi) = \psi \circ i = \varphi$. Therefore, $\varphi \in \operatorname{range} i'$. So, $\operatorname{range} i' = U'$. By Theorem 3.44, $\dim U' = \dim U$. So, $\dim V = \dim U^0 + \dim U$. □

Theorem 3.48 (condition for the annihilator to equal $\{0\}$ or the whole space)**.** *Suppose V is finite-dimensional and U is a subspace of V. Then:*

(a) $U^0 = \{0\}$ iff $U = V$.

(b) $U^0 = V'$ iff $U = \{0\}$.

Proof. We first prove (a). Suppose $U^0 = \{0\}$. By Theorem 3.47, $\dim U^0 = 0$, so $\dim V = \dim U$. Since U is a subspace of V, $U = V$ by Theorem 2.12. Now, suppose $U = V$. Again, we can apply Theorem 3.47 and get that $\dim U^0 = 0$,

so $U^0 = \{0\}$.
Nowe we prove (b). By Theorem 3.44, $\dim V = \dim V'$. Now, suppose $U^0 = V'$. Then, $\dim U^0 = \dim V' = \dim V$, so $\dim U = 0 \implies U = \{0\}$ by Theorem 3.47. Now, suppose $U = \{0\}$. Theorem 3.47 implies that $\dim U^0 = \dim V = \dim V'$. Since U^0 is a subspace of V' (Theorem 3.46), $U^0 = V'$ by Theorem 2.12. □

Theorem 3.49 (the null space of T')**.** *Suppose V and W are finite-dimensional and $T \in \mathcal{L}(V, W)$. Then:*

(a) $\operatorname{null} T' = (\operatorname{range} T)^0$.

(b) $\dim \operatorname{null} T' = \dim \operatorname{null} T + \dim W - \dim V$.

Proof. First, we prove part (a). Suppose that $\varphi \in \operatorname{null} T'$. Then, $T'\varphi = 0$, meaning $\varphi(T) = 0$. That is, for every $v \in V$, $(\varphi \circ T)v = 0$. That is, $\varphi(Tv) = 0$ for all $Tv \in \operatorname{range} T$. Therefore, $\varphi \in (\operatorname{range} T)^0$. Now, suppose $\varphi \in (\operatorname{range} T)^\circ$. Then, $\varphi(Tv) = 0$ for all $Tv \in \operatorname{range} T$. hHat is, $(\varphi \circ T)v = 0$ for all $v \in V$. Since $T'\varphi = \varphi \circ T$, this means $(T'\varphi)v = 0$ for all $v \in V$, i.e, that $T'\varphi = 0$. So, $\varphi \in \operatorname{null} T'$.
Now we prove part (b):

$$\begin{aligned}
\dim \operatorname{null} T' &= \dim(\operatorname{range} T)^0 && \text{part (a)} \\
&= \dim W - \dim \operatorname{range} T && \text{Theorem 3.47} \\
&= \dim W - (\dim V - \dim \operatorname{null} T) && \text{Theorem 3.7} \\
&= \dim \operatorname{null} T + \dim W - \dim V
\end{aligned}$$

□

Remark 3.8. In the preceding theorem, part (a) does not require the assumption V and W are finite-dimensional.

Theorem 3.50 (T surjective is equivalent to T' injective)**.** *Suppose V and W are finite-dimensional and $T \in \mathcal{L}(V, W)$. Then, T is surjective iff T' is injective.*

Proof. Suppose T is surjective. By Theorem 3.49:

$$\begin{aligned}
\dim \operatorname{null} T' &= \dim \operatorname{null} T + \dim W - \dim V \\
&= \dim \operatorname{null} T + \dim \operatorname{range} T - \dim V && T \text{ is surjective} \\
&= \dim V - \dim V && \text{Theorem 3.7} \\
&= 0
\end{aligned}$$

Since $\dim \operatorname{null} T' = 0$, $\operatorname{null} T' = \{0\}$, so T' is injective by Theorem 3.5. Now, suppose T' is injective. By Theorem 3.49:

$$\begin{aligned}
0 &= \dim \operatorname{null} T + \dim W - \dim V \\
&= \dim W - (\dim V - \dim \operatorname{null} T) \\
&= \dim W - \dim \operatorname{range} T && \text{Theorem 3.7}
\end{aligned}$$

So, $\dim \operatorname{range} T = \dim W$, meaning T is surjective. □

Theorem 3.51 (the range of T')**.** *Suppose V and W are finite-dimensional and $T \in \mathcal{L}(V, W)$. Then:*

(a) $\dim \operatorname{range} T = \dim \operatorname{range} T'$

(b) $\operatorname{range} T' = (\operatorname{null} T)^0$

Proof. First, we prove (a). Since $T' \in \mathcal{L}(W', V')$ and $\dim W' = \dim W < \infty$ and $\dim V' = \dim V < \infty$ (Theorem 3.44), we can apply Theorem 3.7:

$$\begin{aligned} \dim W = \dim W' &= \dim \operatorname{null} T' + \dim \operatorname{range} T' \\ &= \dim \operatorname{null} T + \dim W - \dim V + \operatorname{range} T \quad \text{Theorem 3.49} \end{aligned}$$

Subtracting from both sides,

$$\begin{aligned} \dim V - \dim \operatorname{null} T &= \dim \operatorname{range} T' \\ \implies \dim \operatorname{range} T &= \dim \operatorname{range} T' \qquad \text{Theorem 3.7} \end{aligned}$$

For part (b), we note first that $(\operatorname{null} T)^0$ and $\operatorname{range} T'$ are subspaces of V' (Theorem 3.4, Theorem 3.46, Theorem 3.6). Let $T'\varphi \in \operatorname{range} T'$. Then, for all $u \in \operatorname{null} T$, $(T'\varphi)u = (\varphi \circ T)u = \varphi(Tu) = \varphi(0) = 0$. So, $T'\varphi \in (\operatorname{null} T)^0$. Therefore, $\operatorname{range} T' \subset (\operatorname{null} T)^0$. Since they are subspaces of the same space, this means $\operatorname{range} T'$ is a subspace of $(\operatorname{null} T)^0$. Also:

$$\begin{aligned} \dim(\operatorname{null} T)^0 &= \dim V - \dim \operatorname{null} T && \text{Theorem 3.7} \\ &= \dim \operatorname{range} T \\ &= \dim \operatorname{range} T' && \text{by part (a)} \end{aligned}$$

By Theorem 2.12, $\dim(\operatorname{null} T)^0 = \dim \operatorname{range} T'$. □

Theorem 3.52 (T injective is equivalent to T' surjective)**.** *Suppose V and W are finite-dimensional and $T \in \mathcal{L}(V, W)$. Then, T is injective iff T' is surjective.*

Proof. We have:

$$\begin{aligned} \dim V' &= \dim V && \text{Theorem 3.44} \\ &= \dim \operatorname{null} T + \dim \operatorname{range} T && \text{Theorem 3.7} \\ &= \dim \operatorname{null} T + \dim \operatorname{range} T' && \text{Theorem 3.51} \end{aligned}$$

If T is injective, then $\dim \operatorname{null} T = 0$ (Theorem 3.5), so $\dim V' = \dim \operatorname{range} T'$. By Theorem 3.6 and Theorem 2.12, $\operatorname{range} T' = V'$, so T' is surjective. If T' is surjective, then $\dim \operatorname{range} T' = \dim V'$, so $\dim \operatorname{null} T = 0$. By Theorem 3.5, T is injective. □

Theorem 3.53 (matrix of T' is transpose of matrix of T)**.** *Suppose V and W are finite-dimensional and $T \in \mathcal{L}(V,W)$. Then, $\mathcal{M}(T') = (\mathcal{M}(T))^T$.*

Proof. We will write $\mathcal{M}(T') = \mathcal{M}(T', (\psi_1, ..., \psi_m), (\varphi_1, ..., \varphi_n))$ and $\mathcal{M}(T) = \mathcal{M}(T, (v_1, ..., v_n), (w_1, ..., w_m))$. Let $A = \mathcal{M}(T)$ and $B = \mathcal{M}(T')$. Then:

$$T'\psi_j = \sum_{k=1}^{n} B_{k,j}\psi_k \text{ for each } j \in [m]$$
$$T\varphi_k = \sum_{j=1}^{m} A_{j,k}\varphi_j \text{ for each } k \in [n]$$

Then, for each $l \in [n]$:

$$\begin{aligned}(T'\psi_j)v_l &= \sum_{k=1}^{n} B_{k,j}\varphi_k(v_l) \\ &= B_{l,j}\end{aligned}$$

where we use Definition 3.29. Expanding $(T'\psi_j)v_l$:

$$\begin{aligned}(T'\psi_j)v_l &= \psi_j(Tv_l) \\ &= \psi_j\Big(\sum_{r=1}^{m} A_{r,l}w_j\Big) \\ &= A_{j,l}\end{aligned}$$

where we use Definition 3.29. Therefore, $B_{l,j} = A_{j,l}$ for each $j \in [m]$ and $l \in [n]$. Therefore, $(\mathcal{M}(T))^T = \mathcal{M}(T')$. □

We can prove that row rank equals column rank through duality.

Theorem 3.54. *Suppose $A \in \mathbb{F}^{m,n}$. Then the column rank of A equals the row rank of A.*

Proof. Define $T : \mathbb{F}^{n,1} \to \mathbb{F}^{m,1}$ by $Tx = Ax$. Thus, $\mathcal{M}(T) = A$, where $\mathcal{M}(T)$ is computed w.r.t the standard bases of $\mathbb{F}^{n,1}$ and $\mathbb{F}^{m,1}$. Now:

$$\begin{aligned}\text{column rank of } A &= \text{column rank of } \mathcal{M}(T) \\ &= \dim \operatorname{range} T && \text{Theorem 3.31} \\ &= \dim \operatorname{range} T' && \text{Theorem 3.51} \\ &= \text{column rank of } \mathcal{M}(T') && \text{Theorem 3.31} \\ &= \text{column rank of } A^T && \text{Theorem 3.53} \\ &= \text{row rank of } A\end{aligned}$$

□

4 Polynomials

4.1 Basic Properties

Definition 4.1 (zero of polynomial)**.** A number $\lambda \in \mathbb{F}$ is called a *zero* (or *root*) of a polynomial $p \in \mathcal{P}(\mathbb{F})$ if $p(\lambda) = 0$.

Theorem 4.1 (each zero of a polynomial corresponds to a degree-one factor)**.** *Suppose m is a positive integer and $p \in \mathcal{P}(\mathbb{F})$ is a polynomial of degree m. Suppose $\lambda \in \mathbb{F}$. Then, $p(\lambda) = 0$ iff there exists a polynomial $q \in \mathcal{P}(\mathbb{F})$ of degree $m-1$ so that:*

$$p(z) = (z - \lambda)q(z)$$

for every $z \in \mathbb{F}$.

Proof. Suppose $p(\lambda) = 0$. Then, $p(z) = p(z) - p(\lambda) = \sum_{k=0}^{m} a_k(z^k - \lambda^k) = \sum_{k=1}^{m} a_k(z^k - \lambda^k)$. Then, for each $k \in [m]$, we can write $z^k - \lambda^k = (z - k)(\sum_{j=1}^{k} \lambda^{j-1} z^{k-j})$. Therefore, $z^k - \lambda^k$ is equal to $z - \lambda$ times some polynomial of degree $k-1$. Then, p equals $z - \lambda$ times some polynomial of degree $m-1$. For the converse, if $p(z) = (z - \lambda)q(z)$, then $p(\lambda) = (\lambda - \lambda)q(\lambda) = 0$. □

Theorem 4.2 (degree m implies at most m zeros)**.** *Suppose m is a positive integer and $p \in \mathcal{P}(\mathbb{F})$ is a polynomial of degree m. Then, p has at most m zeros in $\mathbb{F}$.*

Proof. Suppose p has a root at $z = \lambda_1$. Then, by Theorem 4.1, $p(z) = (z - \lambda_1)q_1(z)$ where q_1 is a degree $m-1$ polynomial. If $m - 1 = 0$, then we are done. Otherwise, we can apply the preceding step to $q_1(z)$ to obtain $q_1(z) = (z - \lambda_2)q_2(z)$ for some $q_1(\lambda_2) = 0$ and $m-2$ polynomial $q_2(z)$. More generally, at step $k \in [m-1]$, $q_k(z) = (z - \lambda_{k+1})q_{k+1}(z)$ for some $q_k(\lambda_{k+1}) = 0$ and $m-k-1$ degree polynomial $q_{k+1}(z)$. At $k = m-1$, the polynomial is of degree 0, and a degree 0 polynomial has no roots unless it is zero itself. It is also clear that $p(\lambda_k) = 0$ for each $k \in [m]$, so p has at most m roots. □

Theorem 4.3 (division algorithm for polynomials)**.** *Suppose that $p, s \in \mathcal{P}(\mathbb{F})$, with $s \neq 0$. Then there exist unique polynomials $q, r \in \mathcal{P}(\mathbb{F})$ such that $p = sq + r$ and $\deg r < \deg s$.*

Proof. Let $n = \deg p$ and $m = \deg s$. Suppose $n < m$. Then, take $q = 0$ and take $r = p$. Then, $\deg r = \deg p < \deg s$, and $p = sq + r$. Now, suppose $n \geq m$. Then, the list

$$1, z, ..., z^{m-1}, s, zs, ..., z^{n-m}s$$

is linearly independent in $\mathcal{P}_n(\mathbb{F})$ (each polynomial has a different degree). Also, it has length $n - m + m - 1 + 1 = n$, so by Theorem 2.11, it is a basis for $\mathcal{P}_n(\mathbb{F})$.

Then, for any $p \in \mathcal{P}_n(\mathbb{F})$, we can write:

$$\begin{aligned} p &= \sum_{k=0}^{m-1} a_k z^k + \sum_{k=0}^{n-m} b_k z^k s \\ &= \sum_{k=0}^{m-1} a_k z^k + s \sum_{k=0}^{n-m} b_k z^k \end{aligned}$$

for some coefficients $a_k, b_k \in \mathbb{F}$. We can take $q = \sum_{k=1}^{n-m} b_k z^k$ and $r = \sum_{k=0}^{m-1} a_k z^k$ to get $p = sq + r$ with $\deg r < \deg s$. Uniqueness of q and r follows from Theorem 2.5. □

4.2 Factorization Over $\mathbb{C}$

For this section and the next, refer to notes on analysis and complex analysis. The proofs will not be reproduced here.

Theorem 4.4 (fundamental theorem of algebra, first version)**.** *Every nonconstant polynomial with complex coefficients has a zero in* $\mathbb{C}$*.*

Theorem 4.5 (fundamental theorem of algebra, second version)**.** *If* $p \in \mathcal{P}(\mathbb{C})$ *is a nonconstant polynomial, then* p *has a unique factorization (except for the order of the factors) of the form:*

$$p(z) = c(z - \lambda_1)...(z - \lambda_m)$$

where $c, \lambda_1, ..., \lambda_m \in \mathbb{C}$*.*

4.3 Factorization Over $\mathbb{R}$

Theorem 4.6 (polynomial with real coefficients have nonreal zeros in pairs)**.** *Suppose* $p \in \mathcal{P}(\mathbb{C})$ *and has real coefficients. If* $\lambda \in \mathbb{C}$ *is a root for* p*, then* $\overline{\lambda}$ *is a root for* p*.*

Proof. Since $p \in \mathcal{P}(\mathbb{C})$, $p(z) = \sum_{k=0}^{m} a_k z^k$ for some coefficients $a_0, ..., a_m \in \mathbb{C}$. Then:

$$p(\lambda) = \sum_{k=0}^{m} a_k \lambda^k$$

Applying properties of complex conjugates:

$$\begin{aligned} p(\overline{\lambda}) &= \sum_{k=1}^{m} a_k (\overline{\lambda})^k \\ &= \overline{\sum_{k=0}^{m} a_k \lambda^k} \\ &= \overline{0} = 0 \end{aligned}$$

□

Theorem 4.7 (factorization of a quadratic polynomial)**.** *Suppose $b, c \in \mathbb{R}$ Then there is a polynomial factorization of the form $x^2 + bx + c = (x - \lambda_1)(x - \lambda_2)$ with $\lambda_1, \lambda_2 \in \mathbb{R}$ if and only if $b^2 \geq 4c$.*

Theorem 4.8 (factorization of a polynomial over $\mathbb{R}$)**.** *Suppose $p \in \mathcal{P}(\mathbb{R})$ is a nonconstant polynomial. Then p has a unique factorization (except for the order of the factors) of the form: $p(x) = c(x - \lambda_1)...(x - \lambda_m)(x^2 + b_1 x + c_1)...(x^2 + b_M x + c_M)$ where $c, \lambda_1, .., \lambda_m, b_1, ..., b_M, c_1, ..., c_M \in \mathbb{R}$, with $b_k^2 < 4c_k$ for each k. Note either M or m may be zero.*

5 Eigenvalues and Eigenvectors

5.1 Invariant Subspaces

Definition 5.1 (operator)**.** A linear map from a vector space to itself is called an *operator*.

Definition 5.2 (invariant subspace)**.** Suppose $T \in \mathcal{L}(V)$. A subspace U of V is called *invariant* under T if $Tu \in U$ for each $u \in U$.

Example 5.1. Let $T \in \mathcal{L}(V)$. Then, range T, null T, V, and $\{0\}$ are all invariant under T.

Consider the subspace $U = \{\lambda v : \lambda \in \mathbb{F}\}$. If an operator T is invariant under U, then $Tv \in U$, meaning $Tv = \lambda v$ for some $\lambda \in \mathbb{F}$. This motivates the next definition.

Definition 5.3 (eigenvalue)**.** Suppose $T \in \mathcal{L}(V)$. A number $\lambda \in \mathbb{F}$ is calld an *eigenvalue* of T is there exists $v \in V$ such that $v \neq 0$ and $Tv = \lambda v$.

Remark 5.1. T has an eigenvalue if and only if V has a one-dimensional subspace invariant under T.

Theorem 5.1 (equivalent conditions to be an eigenvalue)**.** *Suppose V is finite-dimensional, $T \in \mathcal{L}(V)$, and $\lambda \in \mathbb{F}$. Then the following are equivalent:*

(a) λ is an eigenvalue of T.

(b) $T - \lambda I$ *is not injective.*

(c) $T - \lambda I$ *is not surjective.*

(d) $T - \lambda I$ *is not invertible.*

Proof. If λ is an eigenvalue of T, then $Tv = \lambda v$ for some $v \in V, v \neq 0$. Then, $(T - \lambda I)v = 0$. Therefore, $v \in \text{null}(T - \lambda I)$, so $\text{null}(T - \lambda I) \neq \{0\}$. By Theorem 3.5, $T - \lambda I$ is not injective. By Theorem 3.24, $T - \lambda I$ is not surjective. By Definition 3.15, $T - \lambda I$ is not invertible.
If $T - \lambda I$ is not invertible, then by Theorem 3.24 implies that $T - \lambda I$ is not injective. Then, Theorem 3.5, $(T - \lambda I)v = 0$ for some $v \neq 0$. That is, $Tv = \lambda v$ for some $v \neq 0$. So, T has an eigenvalue. □

Definition 5.4 (eigenvector)**.** Suppose $T \in \mathcal{L}(V)$ and $\lambda \in \mathbb{F}$ is an eigenvalue of T. A vector $v \in V$ is called an *eigenvector* of T corresponding to λ if $v \neq 0$ and $Tv = \lambda v$.

Theorem 5.2 (linearly independent eigenvectors)**.** *Suppose $T \in \mathcal{L}(V)$. Then every list of eigenvectors of T corresponding to distinct eigenvalues of T is linearly independent.*

Proof. Suppose the result is false. Then there exists a smallest positive integer n so that $v_1, ..., v_n$ is a linearly dependent list of eigenvectors of T with corresponding, distinct eigenvalues of T, $\lambda_1, ..., \lambda_n$. Since $v_1 \neq 0$, it follows that $n \geq 2$ for it to be a linearly dependent list. By the minimality of n, there exist coefficients $c_1, ..., c_n \in \mathbb{F}$ so that:

$$\sum_{k=1}^{n} c_k v_k = 0$$

where $c_k \neq 0$ for all $k \in [n]$. Then, applying $(T - \lambda_n I)$ to both sides, we get that:

$$\sum_{k=1}^{n-1} c_k (T - \lambda_n I) v_k = 0$$

which simplifies to:

$$\sum_{k=1}^{n-1} c_k (\lambda_k - \lambda_n) v_k = 0$$

Since the eigenvalues are distinct, this implies that $v_1, ..., v_{n-1}$ is a linearly dependent list of $n - 1$ vectors, contradicting the minimality of n. □

Theorem 5.3 (operator cannot have more eigenvalues than dimension of vector space)**.** *Suppose V is finite-dimensional. Then, each operator on V has at most* $\dim V$ *distinct eigenvalues.*

Proof. By Theorem 5.2, every list of eigenvectors of an operator $T \in \mathcal{L}(V)$ is linearly independent. By Theorem 2.3, the length of any linearly independent list in V cannot exceed that of a spanning list. Since the basis of V is a spanning list with $\dim V$ vectors, the list of eigenvectors of T must have length at most $\dim V$, and so the list of eigenvalues of T must have length at most $\dim V$. $\square$

Definition 5.5 (T^m)**.** Suppose $T \in \mathcal{L}(V)$ and m is a positive integer. Then:

(a) $T^m \in \mathcal{L}(V)$ is defined by $T^m = T \circ T \circ ... \circ T$ (m times)

(b) T^0 is defined to be the identity operator on V

(c) If T is invertible with inverse T^{-1}, then $T^{-m} \in \mathcal{L}(V)$ is defined by $T^{-m} = (T^{-1})^m$.

Definition 5.6 ($p(T)$)**.** Suppose $T \in \mathcal{L}(V)$ and $p \in \mathcal{P}(\mathbb{F})$ is a polynomial given by:

$$p(z) = \sum_{k=0}^{m} a_k z^k$$

for all $z \in \mathbb{F}$. Then, $p(T)$ is the operator on V defined by:

$$p(T) = \sum_{k=0}^{m} a_k T^k$$

Definition 5.7 (product of polynomials)**.** If $p, q \in \mathcal{P}(\mathbb{F})$, then $pq \in \mathcal{P}(\mathbb{F})$ is the polynomial defined by:

$$(pq)(z) = p(z)q(z)$$

for all $z \in \mathbb{F}$.

Theorem 5.4 (multiplicative properties)**.** *Suppose $p, q \in \mathcal{P}(\mathbb{F})$ and $T \in \mathcal{L}(V)$. Then:*

(a) $(pq)(T) = p(T)q(T)$

(b) $p(T)q(T) = q(T)p(T)$

Theorem 5.5 (null space and range of $p(T)$ are invariant under T)**.** *Suppose that $T \in \mathcal{L}(V)$ and $p \in \mathcal{P}(\mathbb{F})$. Then* $\operatorname{null} p(T)$ *and* $\operatorname{range} p(T)$ *are invariant under T.*

Proof. Let $u \in \operatorname{null} p(T)$. Then, $p(T)(Tu) = T(p(T)u) = T(0) = 0$. Let $u \in \operatorname{range} p(T)$. Then, $u = p(T)v$ for some $v \in V$. Then, $Tu = T(p(T)v) = p(T)(Tv) \in \operatorname{range} p(T)$. $\square$

5.2 The Minimal Polynomial

Theorem 5.6 (existence of eigenvalues)**.** *Every operator on a finite-dimensional nonzero complex vector space has an eigenvalue.*

Proof. Let $T \in \mathcal{L}(V)$ and suppose that $v \neq 0$. Then, consider the list $v, Tv, T^2v, ..., T^nv$ where $n = \dim V$. Since this is a list of length $n+1$, $v, Tv, ..., T^nv$ cannot be linearly independent by Theorem 2.3. So, some linear combination of $v, Tv, T^2v, ..., T^nv$ is equal to 0 where not all the coefficients are zero. That is, there exists a smallest positive integer m so that $\sum_{k=0}^{m} c_k T^k v = 0$ for some coefficients $c_0, ..., c_m \in \mathbb{F}$ where $c_k \neq 0$ for all $k \in \{0, ..., m\}$. Then, if we let $p(T) = \sum_{k=0}^{n} c_k T^k$, we get:

$$p(T)v = 0$$

By Theorem 4.4, the polynomial $p(z)$ has a root λ so that $p(\lambda) = 0$. By Theorem 4.1, $p(z) = (z-\lambda)q(z)$ for some polynomial $q(z)$ satisfying $\deg q \leq \deg p$ for all $z \in \mathbb{F}$. Then, $p(T) = (T - \lambda I)q(T)$. Furthermore, $0 = p(T)v = (T-\lambda I)q(T)v$. It cannot be that $q(T)v = 0$ since $p(T)$ is the smallest polynomial for which this is true. So, $(T - \lambda I)v = 0$, so λ is an eigenvalue of T. □

Definition 5.8 (monic polynomial)**.** A *monic polynomial* is a polynomial whose highest degree coefficient equals 1.

Theorem 5.7 (existence, uniqueness, and degree of minimal polynomial)**.** *Suppose V is finite-dimensional and $T \in \mathcal{L}(V)$. Then there is a unique monic polynomial $p \in \mathcal{P}(\mathbb{F})$ of smallest degree such that $p(T) = 0$. Furthermore,* $\deg p \leq \dim V$.

Proof. If $\dim V = 0$, then $V = \{0\}$ and $p(T) = T^0 = I$ is the unique monic polynomial for which $p(T) = 0$ (since $v = 0$ is the only vector and $p(T)v = 0$). Also, $0 = \deg p \leq \dim V = 0$.
Suppose we have proven the result for vector spaces of dimension less than $\dim V = n > 0$. Let $v \in V$ so that $v \neq 0$. The list $v, Tv, ..., T^nv$. This list has length $n+1$ and is therefore linearly dependent.
By Lemma 2.1, there exists a smallest positive integer m so that T^mv is a linear combination of $v, Tv, ..., T^{m-1}v$. Therefore, there exists $c_0, ..., c_{m-1} \in \mathbb{F}$ such that:

$$\sum_{k=0}^{m-1} c_k T^k v + T^m v = 0$$

Define a monic polynomial $q \in \mathcal{P}_m(\mathbb{F})$ by:

$$q(z) = \sum_{k=0}^{m-1} c_k z^k + z^m$$

Then, $q(T)v = 0$. If k is a nonnegative integer, we also have:

$$q(T)(T^k v) = T^k(q(T)v) = T^k(0) = 0$$

by Theorem 5.4. Also, since m is the smallest positive integer so that the list $v, Tv, ..., T^m v$ is linearly dependent, that means the list $v, Tv, ..., T^{m-1}v$ is linearly independent. As shown by the above equation, each $T^k v \in \operatorname{null} q(T)$ for $k \in \{0, ..., m-1\}$. By Theorem 2.3, $\dim \operatorname{null} q(T) \geq m$. By Theorem 3.7:

$$\dim \operatorname{range} q(T) = \dim V - \dim \operatorname{null} q(T) \leq \dim V - m$$

Because $\operatorname{range} q(T)$ is invariant under T by Theorem 5.5, we can apply the induction hypothesis to the operator $T|_{\operatorname{range} q(T)}$ on the vector space $\operatorname{range} q(T)$. Therefore, there exists a monic polynomial $s \in \mathcal{P}(\mathbb{F})$ with:

$$\deg s \leq \dim V - m \text{ and } s(T|_{\operatorname{range} q(T)}) = 0$$

So, for all $v \in V$: $(sq)(T)(v) = s(T)(q(T)v) = 0$ because $q(T)v \in \operatorname{range} q(T)$ and $s(T)|_{\operatorname{range} q(T)} = s(T|_{\operatorname{range} q(T)}) = 0$. Therefore, sq is a monic polynomial such that $\deg sq \leq \dim V$ and $(sq)(T) = 0$.
We have shown that there exists a monic polynomial of degree at most $\dim V$ that when applied to T yields the 0 operator. Since the degrees of a polynomial are nonnegative (and thus well-ordered), there exists a *smallest* degree monic polynomial with this property. This shows existence.
To show uniqueness, let $p \in \mathcal{P}(\mathbb{F})$ be a monic polynomial of smallest degree such that $p(T) = 0$. Suppose $r \in \mathcal{P}(\mathbb{F})$ is a monic polynomial of the same degree as p and $r(T) = 0$. Then, $(p-r)(T) = 0$ and $\deg(p-r) \leq \deg p$. If $p - r$ were not equal to 0, we could divide $p - r$ by the coefficient of the highest order term in $p - r$ to get a monic polynomial of smaller degree than p that when applied to T yields the 0 operator. Therefore, $p - r = 0$, as desired. □

Definition 5.9 (minimal polynomial)**.** Suppose V is finite-dimensional and $T \in \mathcal{L}(V)$. Then the *minimal polynomial* of T is the unique monic polynomial $p \in \mathcal{P}(\mathbb{F})$ of smallest degree such that $p(T) = 0$.

Remark 5.2. To find the minimal polynomial, we must find a polynomial $p(T) = 0$ that is monic. That is, we need:

$$p(T) = c_0 I + c_1 T + ... + c_{m-1} T^{m-1} + T^m = 0$$

That is, we need to find:

$$c_0 I + c_1 T + ... + c_{m-1} T^{m-1} = -T^m$$

Theorem 5.8 (eigenvalues are the zeros of the minimal polynomial)**.** *Suppose V is finite-dimensional and $T \in \mathcal{L}(V)$.*

1. *The zeros of the minimal polynomial of T are the eigenvalues of T*
2. *If V is a complex vector space, then the minimal polynomial of T has the form:*

   $$(z - \lambda_1)...(z - \lambda_m)$$

   where $\lambda_1, ..., \lambda_m$ is a list of all the eigenvalues of T, possibly with repetitions.

Proof. Clearly, part (b) follows from part (a) by Theorem 4.5. Now we prove part (a). First, suppose $\lambda \in \mathbb{F}$ is a zero of p, the minimal polynomial of T. Then, p can be written in the form:

$$p(z) = (z - \lambda)q(z)$$

where q is a monic polynomial with coefficients in $\mathbb{F}$ by Theorem 4.1. Then, since $p(T) = 0$, we have:

$$0 = (T - \lambda I)q(T)$$

Since $\deg q < \deg p$, it follows that there exists $v \in V$ so that $q(T)v \neq 0$. Then, we would have $(T - \lambda I)(q(T)v) = 0$. Therefore, λ is an eigenvalue of T. Now, suppose λ is an eigenvalue of T. Then, $Tv = \lambda v$ for some $v \in V$, $v \neq 0$. Then, $T^k v = \lambda^k v$ for each nonnegative integer k. Therefore, $p(T)v = p(\lambda)v$. Because p is the minimal polynomial of T, we have $p(T)v = 0$. Hence, $p(\lambda) = 0$. Therefore, λ is a zero of p. □

Theorem 5.9 ($q(T) = 0$ iff q is a polynomial multiple of the minimal polynomial)**.** *Suppose V is finite-dimensional, $T \in \mathcal{L}(V)$, and $q \in \mathcal{P}(\mathbb{F})$. Then, $q(T) = 0$ iff $q$is a polynomial multiple of the minimal polynomial of T.*

Proof. Let p be the minimal polynomial of T. Suppose that q is a polynomial multiple of the minimal polynomial of T. Then, $q = sp$ for some polynomial s. Then, $q(T)v = (sp)(T)(v) = (s(T)p(T))v = s(T)(p(T)v) = s(T)(0) = 0$ for each $v \in V$. So, $q(T) = 0$. Now, suppose that $q(T) = 0$. Then, by Theorem 4.3, $q = ps + r$ for some polynomials $s, r \in \mathcal{P}(\mathbb{F})$ with $\deg r < \deg p$. Then, $0 = q(T) = p(T)s(T) + r(T) = r(T)$. Then, $r = 0$, because if $r \neq 0$, we can divide r by its highest degree coefficient to produce a monic polynomial that when applied to T gives 0; this would be a monic polynomial of lower degree than the minimal polynomial, which is a contradiction. Therefore, $q = ps$, so q is a polynomial multiple of p. □

Theorem 5.10 (minimal polynomial of restriction operator)**.** *Suppose V is finite-dimensional and $T \in \mathcal{L}(V)$, and U is a subspace of V that is invariant under T. Then, the minimal polynomial of T is a polynomial multiple of the minimal polynomial of $T|_U$.*

Proof. Let p be the minimal polynomial of T. Since $Tu \in U$ for all $u \in U$, $T|_U \in \mathcal{L}(U)$. Since U is a subspace of V and V is finite-dimensional, Theorem 5.7 implies there exists a minimal polynomial q of $T|_U$. Since $p(T) = 0$, $p(T|_U) = 0$, so Theorem 5.9 implies p is a polynomial multiple of q, as desired. □

Theorem 5.11 (T not invertible iff constant term of minimal polynomial of T is 0)**.** *Suppose V is finite-dimensional and $T \in \mathcal{L}(V)$. Then T is not invertible iff the constant term of the minimal polynomial of T is 0.*

Proof. Let p be the minimal polynomial of T. Suppose T is not invertible. Then, $T - 0I$ is not invertible. By Theorem 5.1, 0 is an eigenvalue of T. By Theorem 5.8, 0 is a root of the minimal polynomial of T, i.e, $p(0) = 0$. That is,

if $p(z) = \sum_{k=0}^{m-1} c_k z^k + z^m$, then, $0 = p(0) = c_0$. These steps can be reversed to show that if the constant term of the minimal polynomial is 0, then T is not invertible. □

Theorem 5.12 (even-dimensional null space)**.** *Suppose $\mathbb{F} = \mathbb{R}$ and V is finite-dimensional. Suppose that $T \in \mathcal{L}(V)$ and $b, c \in \mathbb{R}$ with $b^2 < 4c$. Then, $\dim \operatorname{null}(T^2 + bT + cI)$ is an even number.*

Proof. TODO □

Theorem 5.13 (operators on an odd-dimensional real vector space have eigenvalues)**.** *Every operator on an odd-dimensional real vector space have eigenvalues.*

Proof. TODO □

5.3 Upper-Triangular Matrices

Definition 5.10 (diagonal, upper-triangular matrix)**.** The *diagonal* of a square matrix consists of the entries on the line from the upper-left corner to the bottom-right corner. A square matrix is called *upper-triangular* if all entries below the diagonal are 0.

Theorem 5.14 (conditions for upper-triangular matrix)**.** *Suppose $T \in \mathcal{L}(V)$ and $v_1, ..., v_n$ is a basis for V. Then the following are equivalent:*

(a) The matrix of T with respect to $v_1, ..., v_n$ is upper-triangular.

(b) span $(v_1, ..., v_k)$ is invariant under T for each $k = 1, ..., n$.

(c) $Tv_k \in$ span $(v_1, ..., v_k)$ for each $k = 1, ..., n$.

Proof. First we show that (a) implies (b). The matrix of T with respect to $v_1, ..., v_n$ is defined as:

$$\begin{aligned} Tv_k &= \sum_{j=1}^{n} A_{j,k} v_j \\ &= \sum_{j=1}^{k} A_{j,k} v_j \end{aligned}$$

for each $k \in [n]$ since A is upper-triangular. Let $u \in$ span $(v_1, ..., v_k)$ for some

$k \in [n]$. Then, $u = \sum_{j=1}^{k} c_j v_j$ for some $c_1, ..., c_k \in \mathbb{F}$. Then:

$$\begin{aligned} Tu &= \sum_{j=1}^{k} c_j T v_j \\ &= \sum_{j=1}^{k} \sum_{r=1}^{j} c_j A_{r,j} v_r \\ &= \sum_{j=1}^{k} \sum_{r=j}^{j} c_j A_{r,j} v_r \\ &= \sum_{r=1}^{k} \Big(\sum_{j=r}^{k} c_j A_{r,j} \Big) v_r \end{aligned}$$

So, $Tu \in$ span $(v_1, ..., v_k)$. This proves (b). (b) immediately implies (c) since $v_k \in$ span $(v_1, ..., v_k)$. We show (c) implies (a). By hypothesis, we have:

$$Tv_k = \sum_{j=1}^{k} c_j v_j$$

for some $c_1, ..., c_k \in \mathbb{F}$. The matrix of T with respect to $v_1, ..., v_n$ is:

$$Tv_k = \sum_{j=1}^{n} A_{j,k} v_j$$

Since $v_1, ..., v_n$ is a basis, this implies $\sum_{j=1}^{n} A_{j,k} v_j = \sum_{j=1}^{k} c_j v_j$. Then, $A_{j,k} = 0$ for all $j > k$. That is, the matrix of T is upper-triangular. □

Theorem 5.15 (equation satisfied by operator with upper-triangular matrix)**.** *Suppose $T \in \mathcal{L}(V)$ and V has a basis with respect to which T has an upper-triangular matrix with diagonal entries $\lambda_1, ..., \lambda_n$. Then:*

$$(T - \lambda_1 I)...(T - \lambda_n I) = 0$$

Proof. Let $v_1, ..., v_n$ be a basis for V so that T has an upper triangular matrix with respect to $v_1, ..., v_n$ with diagonal entries $\lambda_1, ..., \lambda_n$. Then, $Tv_1 = A_{1,1} v_1 = \lambda_1 v_1$. So, $(T - \lambda_1 I) v_1 = 0$. By commutativity, $(T - \lambda_1 I)...(T - \lambda_m I) v_1 = 0$ for each $m \in 1, ..., n$. Also, $Tv_2 = A_{1,2} v_1 + \lambda_2 v_2$, so $(T - \lambda_2 I) v_2 \in$ span (v_1). Then, $(T - \lambda_1 I)(T - \lambda_2 I) v_2 = 0$ (by the preceding comments on v_1) and $(T - \lambda_1 I)...(T - \lambda_m I) v_1 = 0$ for each $m = 2, ..., n$. Continuing this pattern, we see that $(T - \lambda_1 I)...(T - \lambda_n I) v_k = 0$ for each $k \in [n]$. Then, $(T - \lambda_1 I)...(T - \lambda_n I) = 0$ since it sends each basis vector to 0. □

Theorem 5.16 (determination of eigenvalues from upper-triangular matrix)**.** *Suppose $T \in \mathcal{L}(V)$ has an upper-triangular matrix with respect to some basis of V. Then, the eigenvalues of T are precisely the entries on the diagonal of that upper-triangular matrix.*

Proof. Let $\lambda_1, ..., \lambda_n$ be the diagonal entries of the matrix of T with respect to some basis of V, $v_1, ..., v_n$. Then, $Tv_1 = \lambda_1 v_1$, so λ_1 is an eigenvalue of T. For each $2 \leq k \leq n$, $(T - \lambda_k I)v_k \in \text{span } (v_1, ..., v_{k-1})$. So, $T - \lambda_k I$ maps $\text{span}(v_1, ..., v_k)$ into $\text{span}(v_1, ..., v_{k-1})$. By Theorem 3.8, $T - \lambda_k I$ is not injective. Then, λ_k is an eigenvalue of T by Theorem 5.1. Let $q(z) = (z - \lambda_1)...(z - \lambda_n)$. By Theorem 5.15, $q(T) = 0$. By Theorem 5.9, q is a polynomial multiple of the minimal polynomial of T. By Theorem 5.8, every zero of the minimal polynomial of T is an eigenvalue of T. Also, every zero of the minimal polynomial of T is a zero of q. So, every eigenvalue of T is a zero of q. This shows that $\lambda_1, ..., \lambda_n$ are all the eigenvalues of T. □

Theorem 5.17 (necessary and sufficient condition to have an upper-triangular matrix)**.** *Suppose V is finite-dimensional and $T \in \mathcal{L}(V)$. Then, T has an upper-triangular matrix with respect to some basis of V if and only if the minimal polynomial of T equals $(z - \lambda_1)...(z - \lambda_m)$ for some $\lambda_1, ..., \lambda_m$.*

Proof. Suppose T has an upper-triangular matrix with respect to some basis of V. Let $\alpha_1, ..., \alpha_n$ denote the diagonal entries of this matrix. Define $q(z) = (z - \alpha_1)...(z - \alpha_n)$. Then, $q(T) = 0$ by Theorem 5.15. By Theorem 5.9, q is a multiple of the minimal polynomial p. Then, the minimal polynomial $p(z) = (z - \lambda_1)...(z - \lambda_m)$ where $\{\lambda_1, ..., \lambda_m\} \subset \{\alpha_1, ..., \alpha_n\}$.
Now suppose that the minimal polynomial of T equals $(z - \lambda_1)...(z - \lambda_m) = p(z)$ for some $\lambda_1, ..., \lambda_m \in \mathbb{F}$. We use induction on m. If $m = 1$, then $z - \lambda_1$ is the minimal polynomial, meaning $(T - \lambda_1 I)v = 0$ for all $v \in V$. That is, $T = \lambda_1 I$, so T has an upper-triangular matrix with respect to some basis of V.
Now, suppose $m > 1$ and the desired result holds for all smaller positive integers. Let $U = \text{range}(T - \lambda_m I)$. Then, U is invariant under T by Theorem 5.5. Therefore, $T|_U$ is an operator on U. If $u \in U$, then $u = (T - \lambda_m I)v$ for some $v \in V$ and:

$$(T - \lambda_1 I)...(T - \lambda_{m-1} I)u = (T - \lambda_1 I)...(T - \lambda_m I)v = 0$$

So, $(z - \lambda_1)...(z - \lambda_{m-1})$ is a multiple of the minimal polynomial of $T|_U$ by Theorem 5.9. So, the minimal polynomial of $T|_U$ is a product of at most $m - 1$ terms of the form $z - \lambda_k$. By the induction hypothesis, there is a basis of $u_1, ..., u_M$ of U so that T has an upper-triangular matrix with respect to $u_1, ..., u_M$. Then, $Tu_k = (T|_U)u_k \in \text{span } (u_1, ..., u_k)$ for each $k \in [M]$ by Theorem 5.14. Extend $u_1, ..., u_m$ to a basis $u_1, ..., u_M, v_1, ..., v_N$ of V. For each $k \in \{1, ..., N\}$, we have $Tv_k = (T - \lambda_k I)v_k + \lambda_k v_k$. Then, since $(T - \lambda_k I)v_k \in U = \text{span}(u_1, ..., u_M)$, it follows $Tv_k \in \text{span}(u_1, ..., u_M, v_1, ..., v_k)$ for each $k \in [M]$. Then, Theorem 5.14 implies that T has an upper-triangular matrix with respect to the basis $u_1, ..., u_M, v_1, ..., v_N$. □

Theorem 5.18 (if $\mathbb{F} = \mathbb{C}$, then every operator on V has an upper-triangular matrix)**.** *Suppose V is a finite-dimensional complex vector space and $T \in \mathcal{L}(V)$. Then T has an upper-triangular matrix with respect to some basis of V.*

Proof. The result follows from Theorem 4.5 and Theorem 5.17. □

5.4 Diagonalizable Operators

Definition 5.11 (diagonal matrix)**.** A *diagonal matrix* is a square matrix that is 0 everywhere except possibly on the diagonal.

Definition 5.12 (diagonalizable)**.** An operator on V is called *diagonalizable* if the operator has a diagonal matrix w.r.t some basis of V.

Definition 5.13 (eigenspace)**.** Suppose $T \in \mathcal{L}(V)$ and $\lambda \in \mathbb{F}$. The *eigenspace* of T corresponding to λ is the subspace $E(\lambda, T)$ of V defined by

$$E(\lambda, T) = \text{null}(T - \lambda I) = \{v \in V : Tv = \lambda v\}$$

Hence, $E(\lambda, T)$ is the set of all eigenvectors of T corresponding to λ along with the 0 vector.

Remark 5.3. We list some things we can immediately conclude from the definitions:

(a) Every diagonal matrix is upper-triangular, so the entries on the diagonal are the eigenvalues.

(b) λ is an eigenvalue of T iff $E(\lambda, T) \neq \{0\}$

(c) If λ is an eigenvalue of $T \in \mathcal{L}(V)$, then T restricted to $E(\lambda, T)$ is just the operator of multiplication by λ.

Theorem 5.19 (sum of eigenspaces is a direct sum)**.** *Suppose $T \in \mathcal{L}(V)$ and $\lambda_1, ..., \lambda_m$ are distinct eigenvalues of T. Then:*

$$E(\lambda_1, T) + ... + E(\lambda_m, T)$$

is a direct sum. Furthermore, if V is finite-dimensional, then:

$$\dim E(\lambda_1, T) + ... + \dim E(\lambda_m, T) \leq \dim V$$

Proof. □